

Nursing Intervention of Patients with Respiratory system Disorders

BY Yihun Miskir

Learning objective

At the end of this session, students will be able to

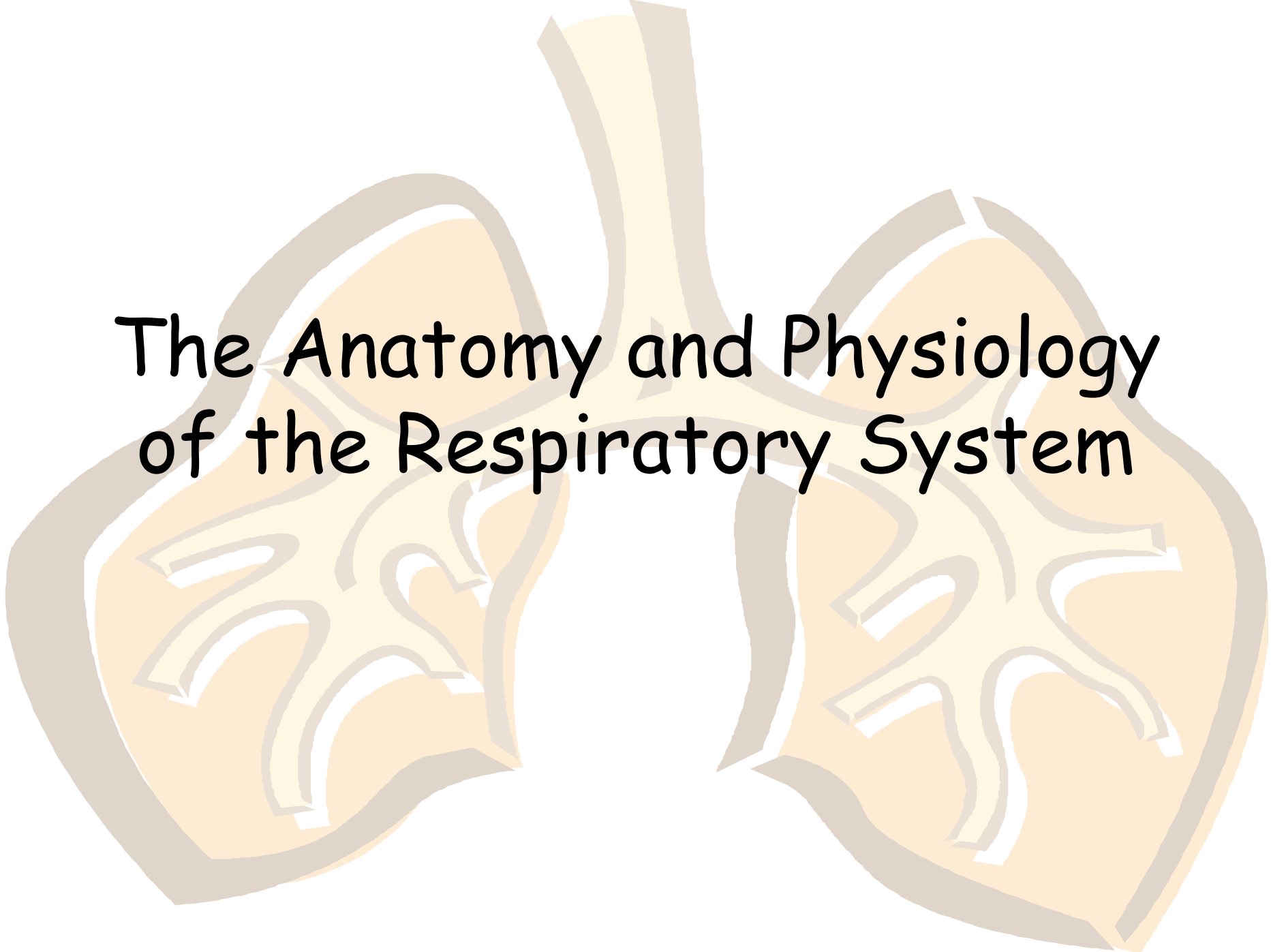
- Describe anatomy & physiology of the RS
- Discuss the chest assessment techniques
- List respiratory system disorders
- Define respiratory system disorders
- Identify causative/risk/contributing factors of RS disorders
- Explain pathophysiologic process of RS
- Describe the clinical manifestations of patients RS

Learning objective....

- Discuss diagnostic procedures used in the diagnosis of RS disorders
- Explain the medical managements of patients with RS disorders
- Develop nursing process in provision of nursing care for patients with RS disorders

Review of anatomy and physiology of RS

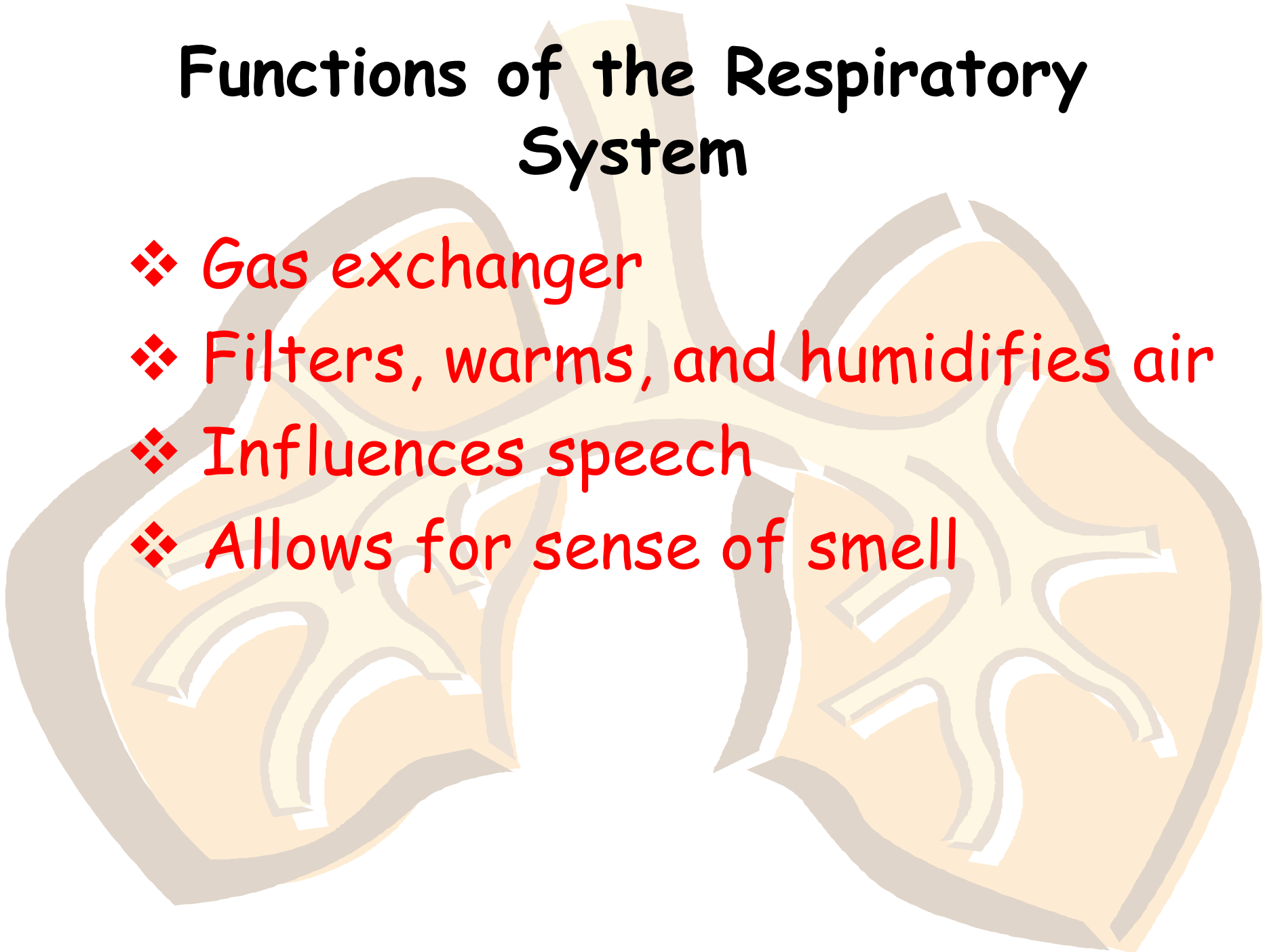
- Discuss the anatomy of respiratory system
- Explain the physiology of RS
- List common respiratory system disorders
- Discuss the common manifestation of RS disorder

A stylized, cartoonish illustration of the human respiratory system. It features two lungs, each with a complex internal branching pattern of bronchi and bronchioles, colored in shades of orange and yellow. A central trachea connects the two lungs. The entire illustration is set against a plain white background.

The Anatomy and Physiology of the Respiratory System

Functions of the Respiratory System

- ❖ Gas exchanger
- ❖ Filters, warms, and humidifies air
- ❖ Influences speech
- ❖ Allows for sense of smell



Divisions of the Respiratory System

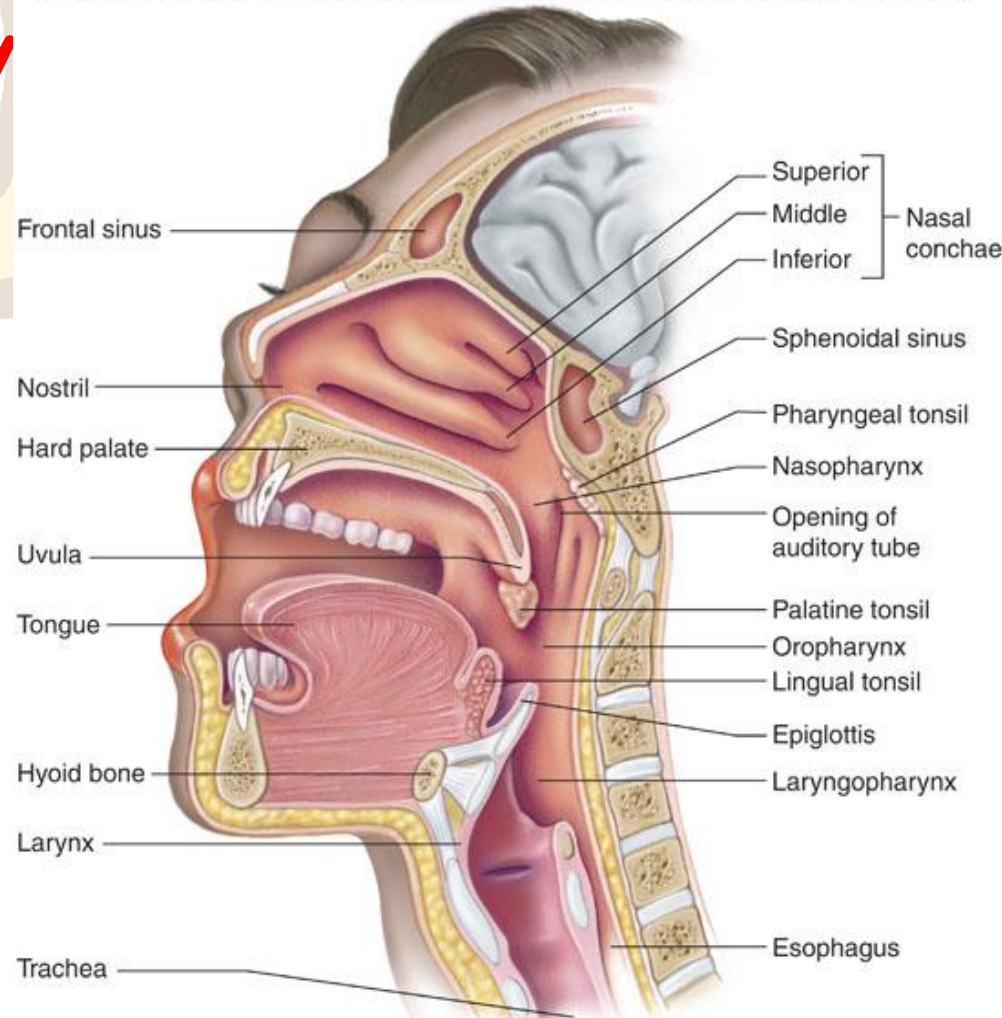
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□ Upper respiratory tract (outside thorax)

❖ Nose

❖ Pharynx

❖ Larynx (voice box)

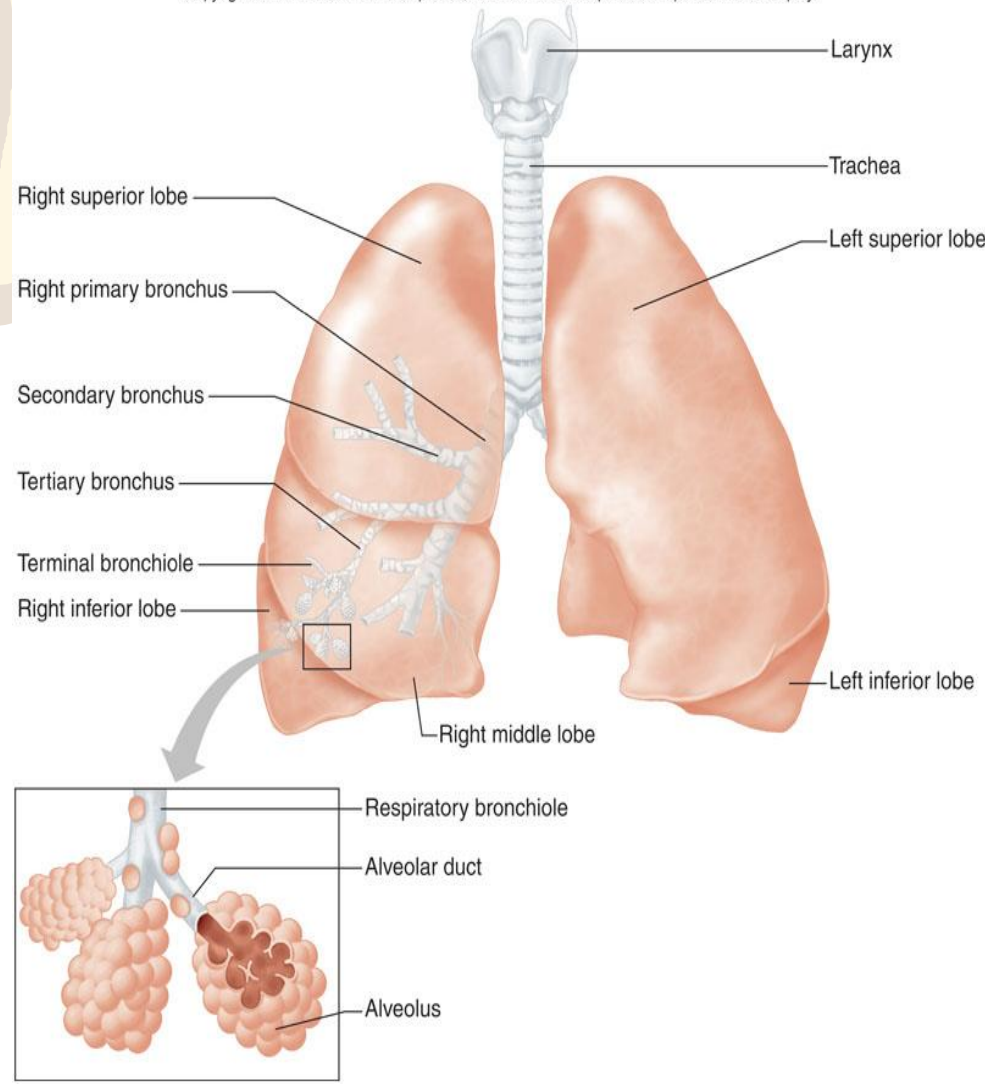


Divisions of the Respiratory System

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❑ Lower respiratory tract (within thorax)

- ❖ Trachea (wind pipe)
- ❖ Bronchial Tree
- ❖ Lungs

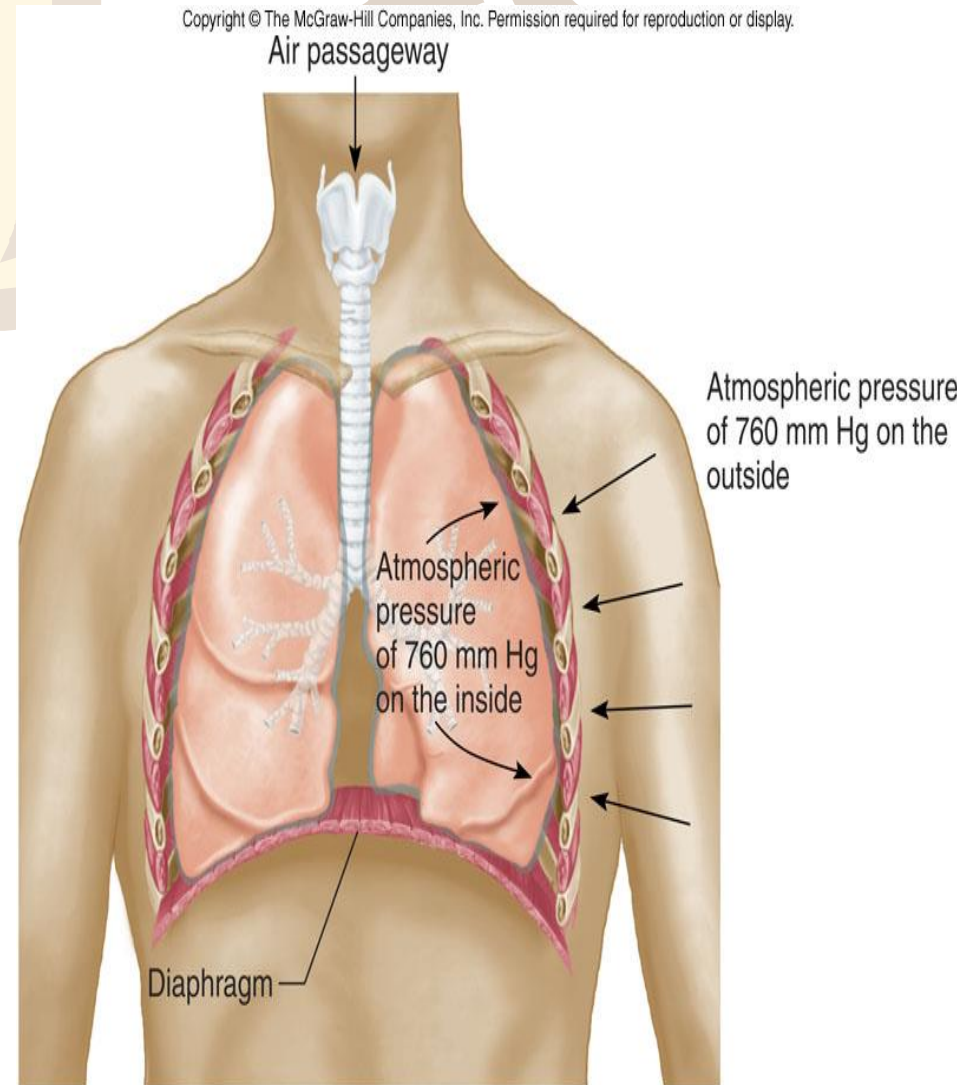


Respiratory Physiology

- **Pulmonary Ventilation = breathing**

- **Mechanism**

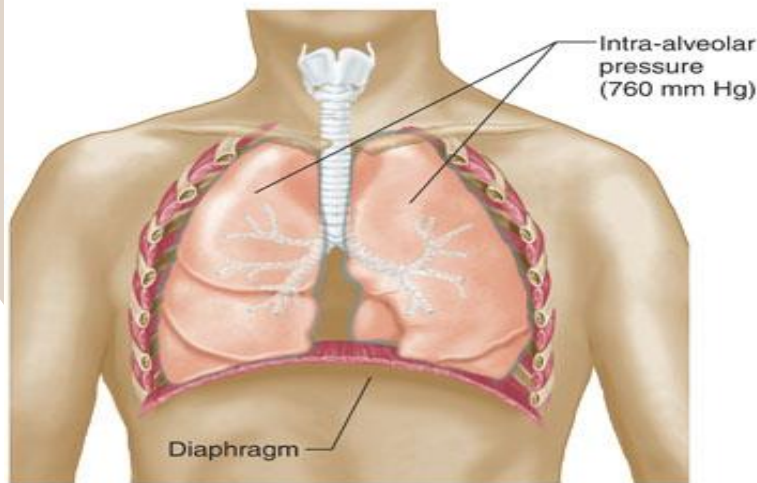
- Movement of gases through a pressure gradient - hi to low.
- When atmospheric pressure (760 mmHg) is greater than lung pressure ---- air flows in = **inspiration**.
- When lung pressure is greater than atmospheric pressure ---- air flows out = **expiration**.



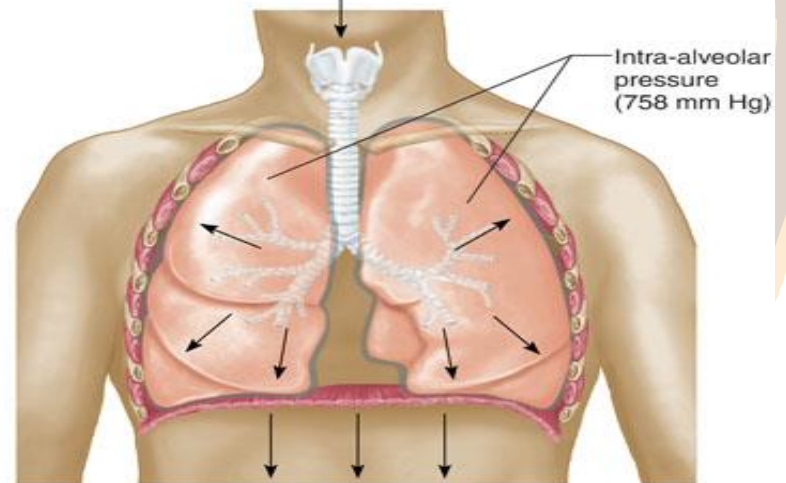
Respiratory Physiology

- Pressure gradients are established by changes in thoracic cavity.
 - increase size in thorax = a decrease in pressure --- air moves in.
 - Decrease size in thorax = increase in pressure --- air moves out.

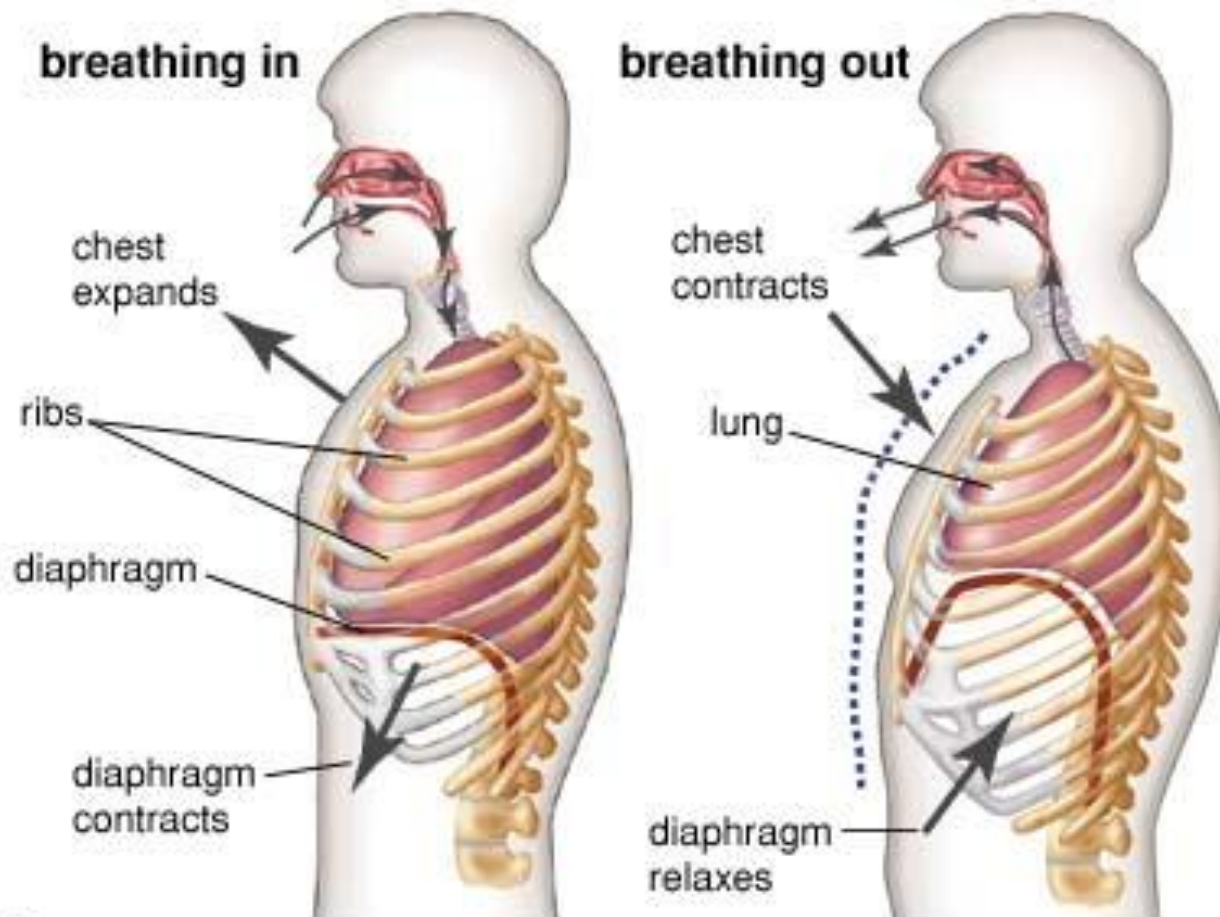
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Atmospheric pressure (760 mm Hg)



(a)



(b)



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Volumes of Air Exchange

- ***Tidal volume*** - amount of air exhaled normally after a typical inspiration. Normal - about 500 ml
- ***Expiratory Reserve volume*** - additional amount of air forcibly expired after tidal expiration (1000 - 1200 ml).
- ***Inspiratory Reserve volume*** - (deep breath) amount of air that can be forcibly inhaled over and above normal.(1900-3300)
- ***Residual volume*** - amount of air that stays trapped in the alveoli (about 1.2 liters).

Volumes of Air Exchange

- ***Vital capacity*** - the largest volume of air an individual can move in and out of the lungs.
- ***Vital capacity = sum of IRV+TV+ERV***
- **Depends of many factors**
 - size of thoracic cavity
 - posture
 - volume of blood in lungs → congestive heart failure, emphysema, disease, etc...

Volumes of Air Exchange

- **Eupnea** - normal quiet breathing, 12-17 breaths per minute.
- **Hyperpnea** - increase in breathing to meet an increased demand by body for oxygen.
- **Hyperventilation** - increase in pulmonary ventilation in excess of the need for oxygen.
 - Someone hysterical
 - exertion
- **Hypoventilation** - decrease in pulmonary ventilation.
- **Apnea** - temporary cessation of breathing at the end of normal expiration.

Breathe into
paper bag.

Review Questions

1 Which of the following is not a function of the respiratory system?

- A. influence speech
- B. Distribution of oxygen to cells
- C. Filtration of air
- D. Warming of air

B

2

The common name for the trachea is _____

Windpipe

3

- The structure known as the Adam's Apple located in neck is the _____

Thyroid Cartilage

4

- The smallest branches of the bronchial tree are the
 - a. Primary bronchi
 - b. Secondary bronchi
 - c. Tertiary bronchi
 - d. Bronchioles

D

5

- The flap or opening to the trachea is the
 - a. Larynx
 - b. Pharynx
 - c. Epiglottis
 - d. Vocal cords

6

- The structure that separates the right and left nasal cavities is the _____

Septum

7

- The incorrect formation of the palantine bone during gestation is known as a _____

Cleft Palate

8

During inspiration which of the following does not occur?

A. Diaphragm contracts

B. Intercostals relax

C. Diaphragm flattens

D. Size of thorax increases

B

9

Which of the following activities is the best analogy of respiration?

A.Exchanging gifts

B.Giving a gift

C.Receiving a gift

D.Sitting in a chair

A

10

Air is forced into the lungs by the contraction of the...

A.Alveoli

B.Bronchioles

C.Diaphragm

D.Heart

C

Assessment of respiratory system

Assessment of respiratory system

❖ Common symptoms

- ✓ Cough
- ✓ Expectoration (sputum)
- ✓ Hemoptysis
- ✓ Chest pain
- ✓ Dyspnea, Shortness of breath, orthopnea, paroxysmal nocturnal dyspnea
- ✓ Wheezes

Assessment....

- Perform the assessment anterior /posterior/ lateral,
- Apex to base, comparing side with side.
- Begin with inspection, then palpate, percuss, and finally auscultate the lungs.

Examination of the Posterior Chest

- **INSPECTION**

- ✓ From a midline position behind the patient, note the *shape of the chest and the way in which it moves*
- ✓ Deformities or asymmetry
- ✓ Abnormal retraction of the interspaces during inspiration

Abnormal posterior chest

- **Kyphosis:** accentuated thoracic curve/hunchback/ may impended respiratory muscles
- **Scoliosis:** lateral 'S' deviation/curvature of the spine.
- **Lordosis:** accentuated lumbar curve or a forward curvature of the lumbar spine associated with kyphosis.

Abnormal posterior chest



Normal spine



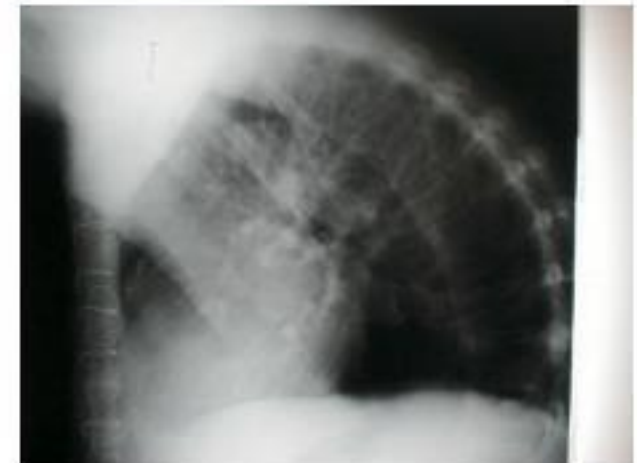
Kyphotic spine

ADAM

Pathologic Changes In Shape Of Spine



Scoliosis (curved to one side)



Thoracic Kyphosis (bent forward)

Abnormal posterior chest

Normal spine



Lordosis of the spine



Exaggerated
lumbar
curve

Abnormal posterior chest



- **PALPATION**

- ✓ *Identify tender areas*

- ✓ *Test chest expansion*



- ✓ Causes of unilateral decrease or delay in chest expansion include
 - chronic fibrotic disease of the underlying lung or pleura,
 - Pleural effusion,
 - Lobar pneumonia

Feel for tactile fremitus

- ✓ Fremitus refers to the palpable vibrations transmitted through the bronchopulmonary tree to the chest wall when the patient speaks
- ✓ Ask the patient to repeat the words “ninety-nine”
- ✓ Fremitus is decreased or absent when the voice is soft or when the transmission of vibrations from the larynx to the surface of the chest is impeded

- Fremitus is typically more prominent in the interscapular area than in the lower lung fields
- More prominent on the right side than on the left.

Percussion

- Percussion helps you establish whether the underlying tissues are air-filled, fluid-filled, or solid

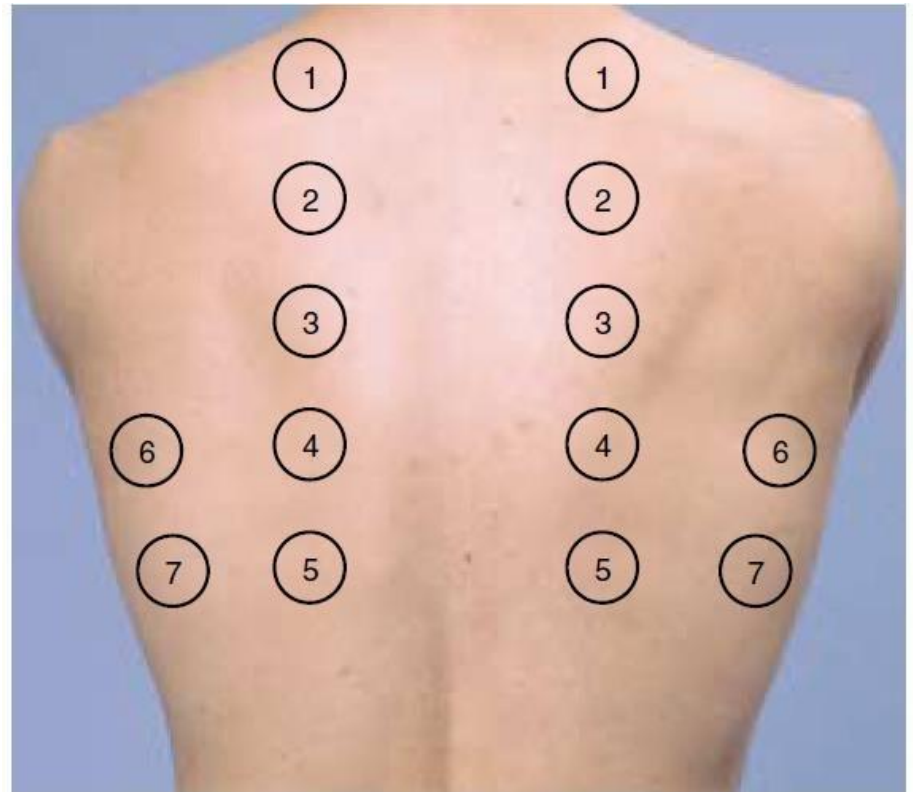


Percussion...

❖ Note

- ✓ Press its distal interphalangeal joint firmly on the surface to be percussed
- ✓ The thumb, 2nd, 4th, and 5th fingers are not touching the chest
- ✓ A thick chest wall requires heavier percussion than a thin one
- ✓ Percuss or strike twice in each location

- Normal lungs are resonant
- Abnormal Finding
 - ✓ Dullness
 - ✓ Hyperresonance

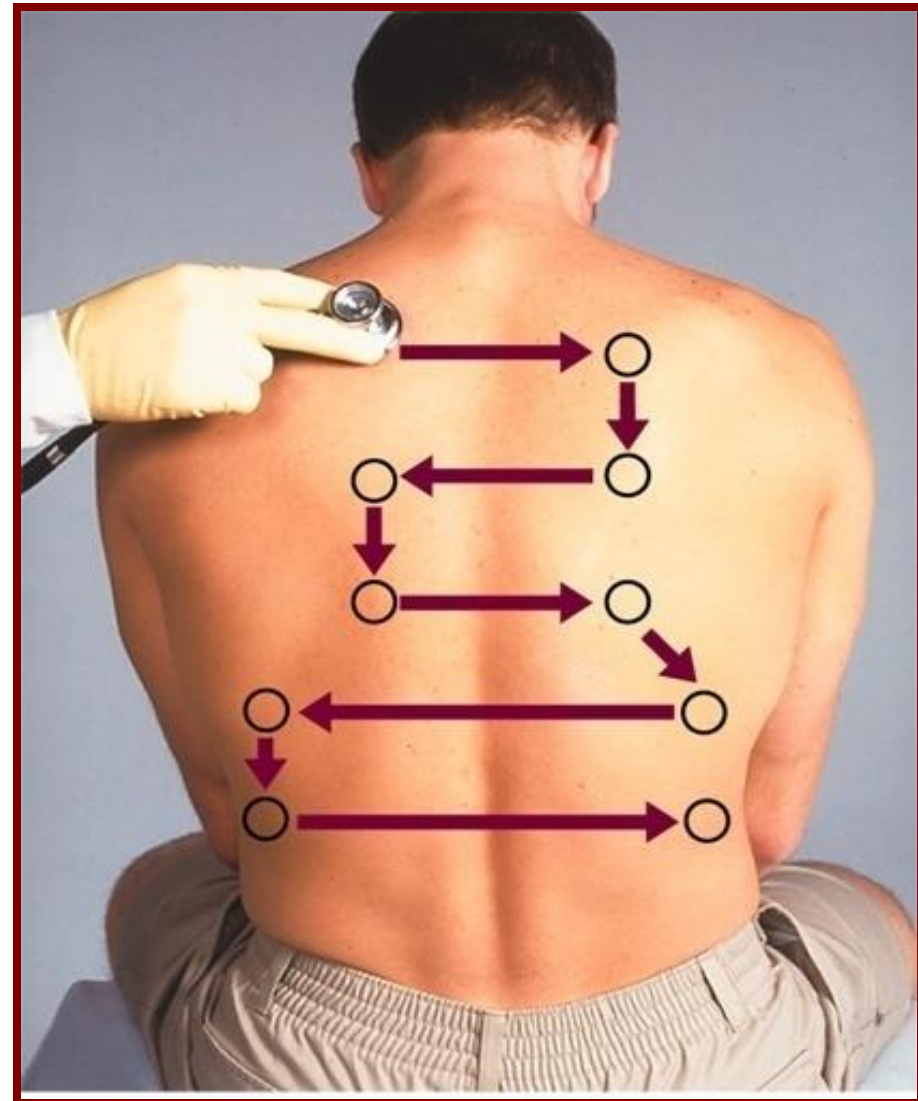
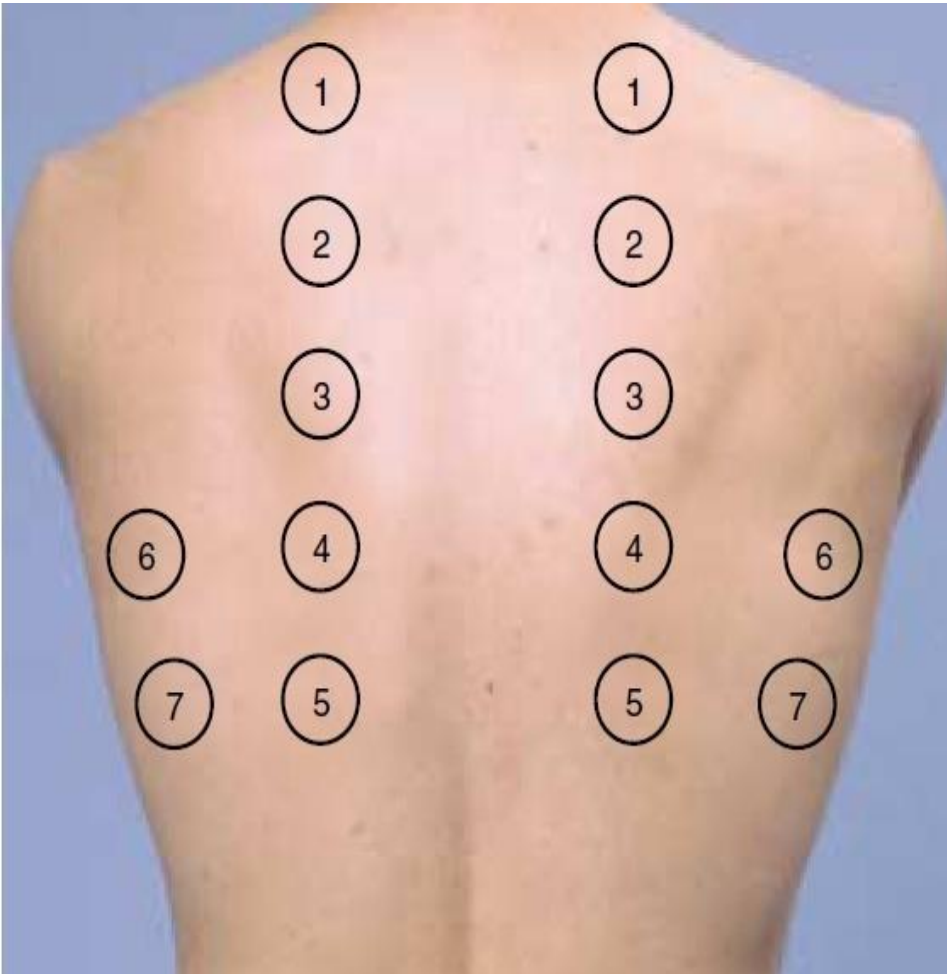


LOCATIONS FOR PERCUSSION AND AUSCULTATION

- ✓ Dullness replaces resonance when fluid or solid tissue replaces air-containing lung
- ✓ Hyperresonance may be heard over the hyperinflated lungs of emphysema or asthma

AUSCULTATION

- The most important examining technique for assessing air flow through the tracheobronchial tree



Abnormal Breath Sounds

- ❖ **Rales (Crackles):-** are discontinuous or intermittent, nonmusical, and brief
 - Sounds like hair being rubbed together
 - Sound produced by **air passing through fluid in air spaces** (CHF, pneumonia).
 - **Usually on inspiration / not expiration**

Abnormal Breath Sounds

❖ **Rhonchi:** continuous **snoring** sound.

- Low pitched (around 200 Hz or lower), snoring quality
- Greater pronounced during expiration.
- It suggest secretions in large airways.
- Etiology: **larger airways are obstructed with mucus or tumor.**

Abnormal Breath Sounds

- ❖ **Wheezing:** high pitched, continuous musical whistling sounds.
- Produced by narrowed airway.
- Can occur during inspiration or expiration.
- Wheezes suggest **narrowed airways**, as in asthma, COPD, or bronchitis.

Abnormal Breath Sounds

❖ **Friction (Pleural) Rub:** Course, dry, grating sound heard **only** with stethoscope.

E.g. Inflamed pleural surfaces rub.

- Sounds similar to cupping hand over ear, scratching back of hand with other hand.
- Usually heard **anteriolateral chest wall**
- Continuous during inspiration and expiration.

Abnormal Breath Sounds

❖ **Stridor:** increased musical wheeze heard over trachea on inspiration.

E.g. Laryngeal obstruction

Transmitted Voice Sounds

- ❖ If you hear abnormally located **broncho-vesicular** or bronchial breath sounds, continue on to assess transmitted voice sounds
- ❖ **Increased transmission of voice sounds** suggests that **air-filled lung has become airless**
- ❖ With a stethoscope, listen in symmetric areas over the chest wall as you do the following:

Bronchophony

- Ask the patient to say 'ninety nine.' Normally the sounds transmitted through the chest wall are muffled and indistinct.
- Louder, clear voice sounds are called bronchophony

Egophony

- ✓ Ask the patient to say 'ee' you will normally hear a muffled long E sound.
- ✓ When 'ee' is heard as 'ay' an E to A change egophony is present

Examination of the Anterior Chest

- **INSPECTION**

- ✓ Observe *the shape of the patient's chest and the movement of the chest wall.*

- Note:

- ✓ Deformities or asymmetry

- ✓ Abnormal retraction of the lower interspaces
during inspiration

- ✓ Local lag or impairment in respiratory movement

Anterior Chest

- **Inspect** respiratory rate, rhythm, depth, and symmetry of chest movements.
- Inspect anteroposterior (AP) to lateral ratio, costal angle, spinal deformities, and condition of skin.
- **Shape and symmetry: Normal adult chest has AP-to-lateral ratio of approximately 1:2 and costal angle less than 90 degrees.**
- Ribs slope obliquely, chest symmetrical in appearance with symmetrical rise and fall when breathing, Skin intact.

- No sternal or intercostal retraction or bulging, unless associated with strenuous activity.
- Musculoskeletal changes associated with aging can increase the AP diameter, giving chest a barrel appearance.
- AP diameter greater in infants (apple-shaped chest).
- **Condition of chest skin: Skin color and hair distribution should be consistent**

Abnormal Anterior Chest

📌 **Barrel chest**-enlarged and rounded cross section to chest associated with COPD

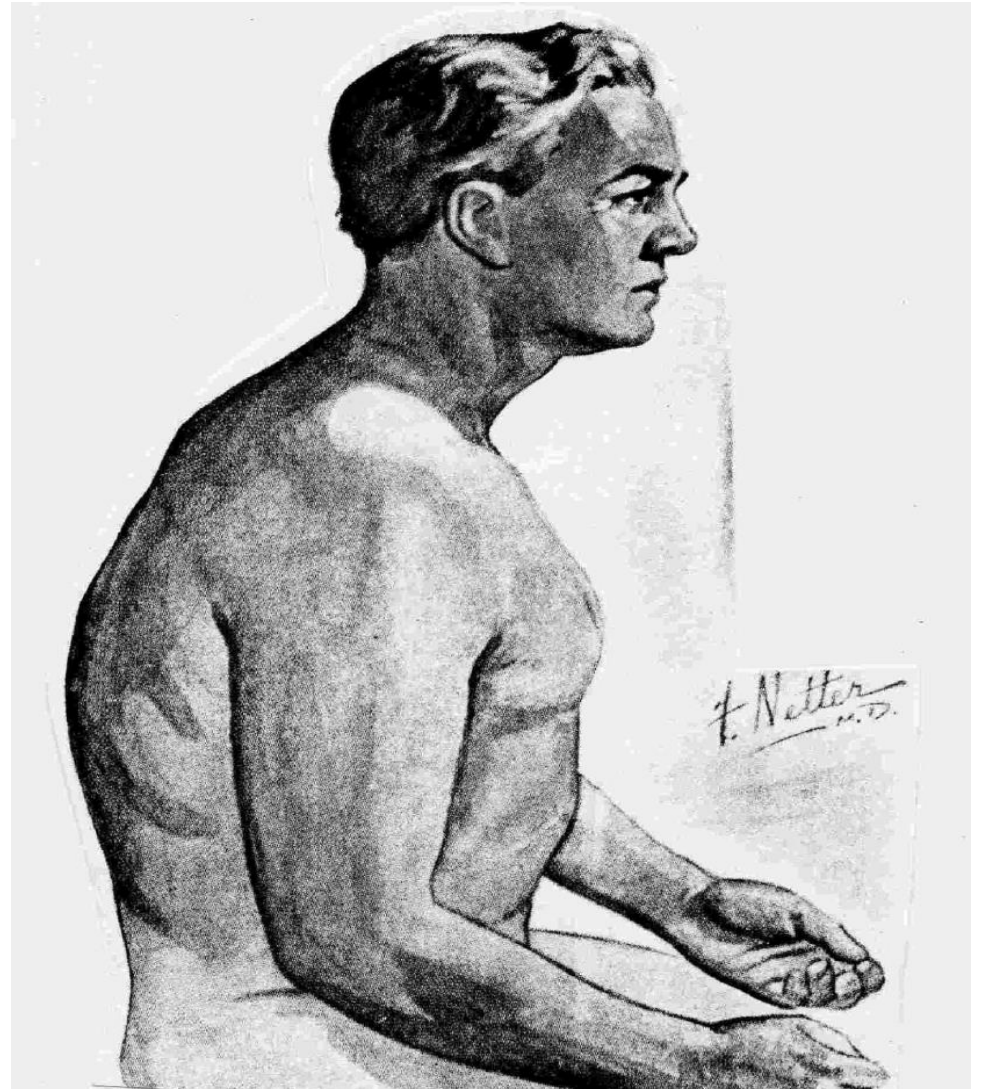
➤ An increased antero-posterior diameter

📌 **Funnel chest**-compression of the lower part of the sternum/**pectus excavatum**/sunken

📌 **Pigeon chest**-characterized by a protruding sternum /**pectus carinatum**/

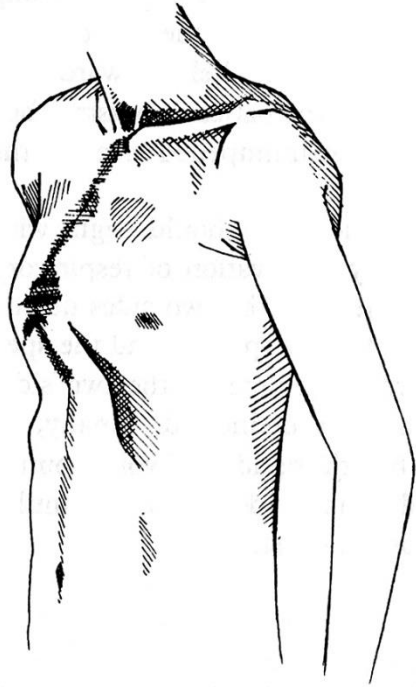
Abnormal anterior chest

Barrel chest

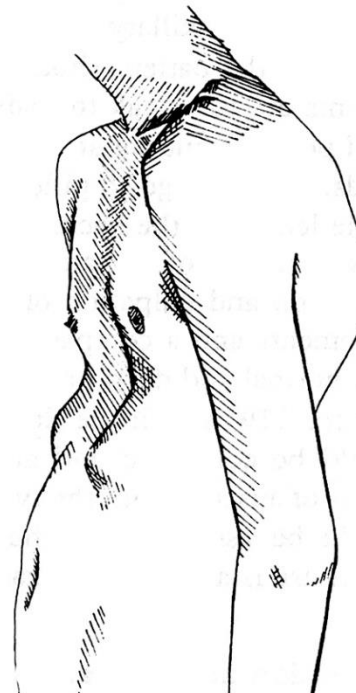


Pathological Change

Pigeon chest



Funnel chest



PALPATION

- Palpation has four potential uses:
 - ✓ *Identification of tender areas*
 - ✓ *Assessment of observed abnormalities*
 - ✓ *Further assessment of chest expansion*
 - ✓ *Assessment of tactile fremitus*

- ✓ Fremitus is usually decreased or absent over the precordium.
- ✓ When examining a woman, gently displace the breasts



PERCUSSION

- Percuss the anterior and lateral chest, again comparing both sides
- Dullness replaces resonance when fluid or solid tissue replaces air-containing lung

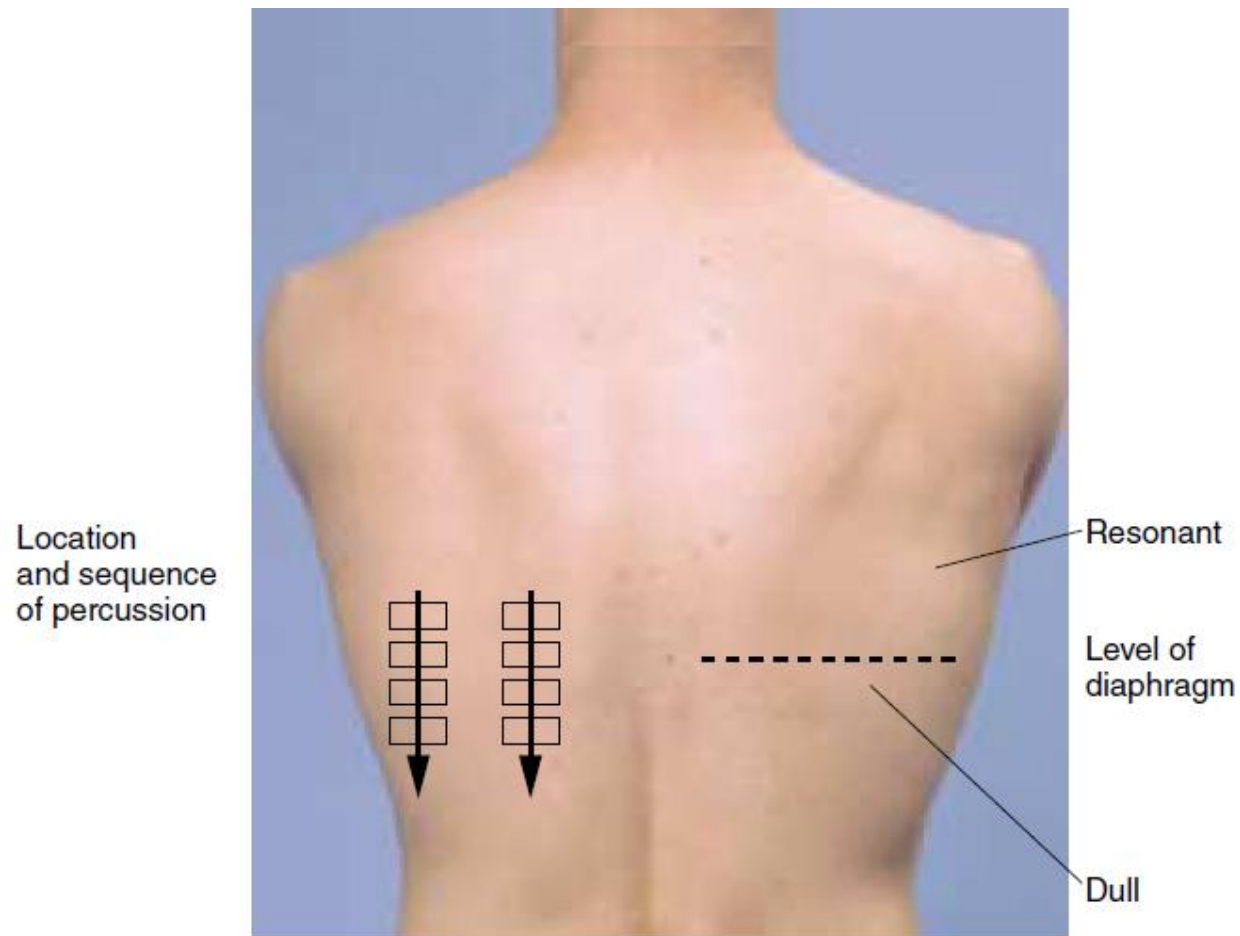


- Resonance to second intercostal space on left; slight dullness over third through fifth intercostal space over heart.
- Resonance to fourth intercostal space on right with dullness from approximately fifth to just above costal margin over liver.
- It will vary with person's size and size of liver.
- **Dullness**: Seen with exudate, fluid, tumors, pneumonia, pulmonary edema, pleural effusion.
- **Hyperresonance**: Noted with air trapping of emphysema.

Diaphragmatic Excursion

- Have patient take a deep breath and fully exhale, then percuss the level of the diaphragm and mark.
- Have the patient take a deep breath and hold it, then percuss the level of the diaphragm.
- Measure the distance between the two marks.
- *3–6 cm diaphragmatic excursion.*
- Decreased diaphragmatic excursion unilaterally or bilaterally: Paralyzed diaphragm, atelectasis, COPD with overinflated lungs.

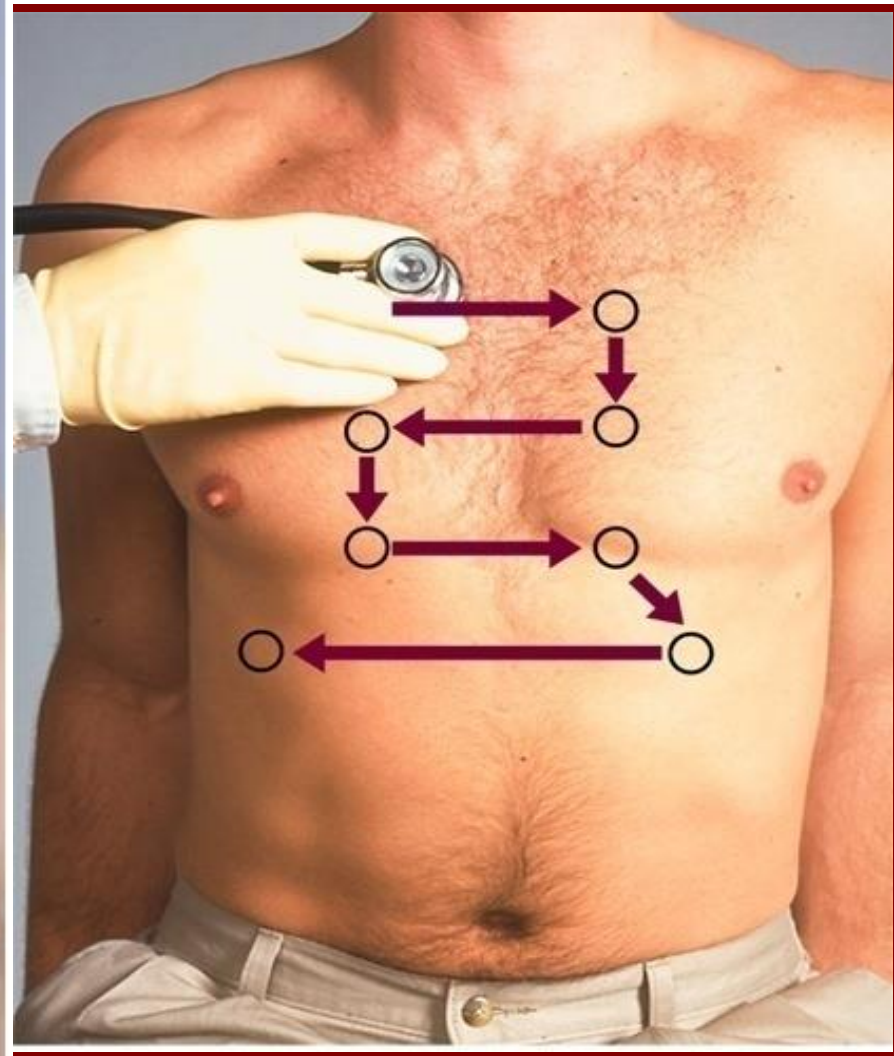
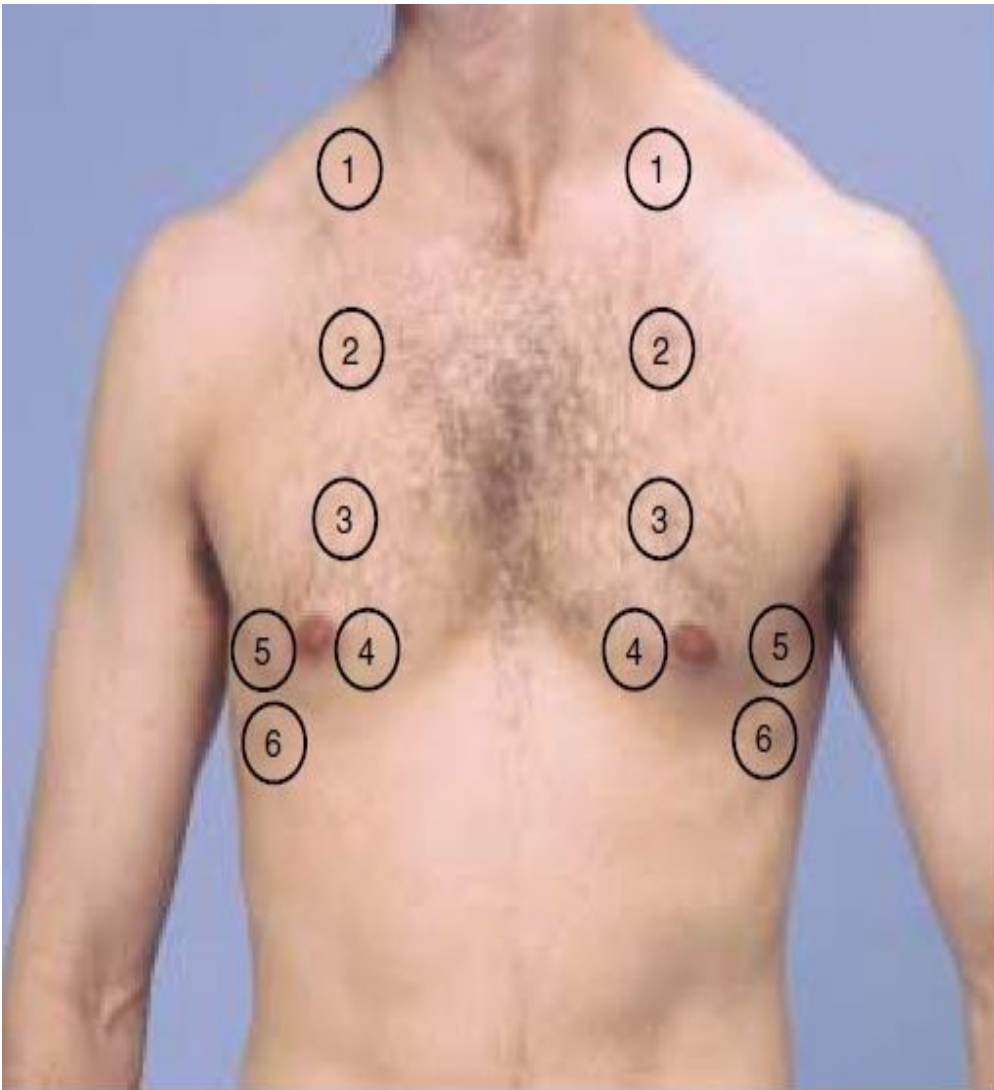
Diaphragmatic Excursion



AUSCULTATION

- Listen to the breath sounds
- Identify any adventitious sounds

Anterior Auscultation



LOCATIONS FOR PERCUSSION AND AUSCULTATION

AUSCULTATING THE LUNGS

Testing for Normal Breath Sounds in Appropriate Locations

- Using the diaphragm portion of the stethoscope, auscultate over auscultatory sites anterior, posterior, and lateral.
- Listen to one full respiratory cycle at each site.
- **Note normal, abnormal, and adventitious sounds.**
- If adventitious sound heard, have patient cough then listen to see if sound has cleared.

Ascultation

Listening breath sound

- **Vesicular** – soft and low pitched heard in inspiration
- **Bronchovesicular**- detected mainly in expiration
- **Bronchial** – louder and high pitched sound

Adventitious sound / added sound/ intermittent nonmusical

- Fine crackles – high pitched
- Course crackle- low pitched

Continuous sound / musical

- wheezes –musical / high pitched /
- Rhonchi- low pitched

- *Bronchial breath* sounds are loud, high-pitched, and hollow, with a short inspiratory phase and a long expiratory phase.
- *They are normally heard in anterior neck and nape of neck posteriorly.*
- Bronchovesicular breath sounds are moderate sounding and medium pitched, with equal inspiratory and expiratory phases.

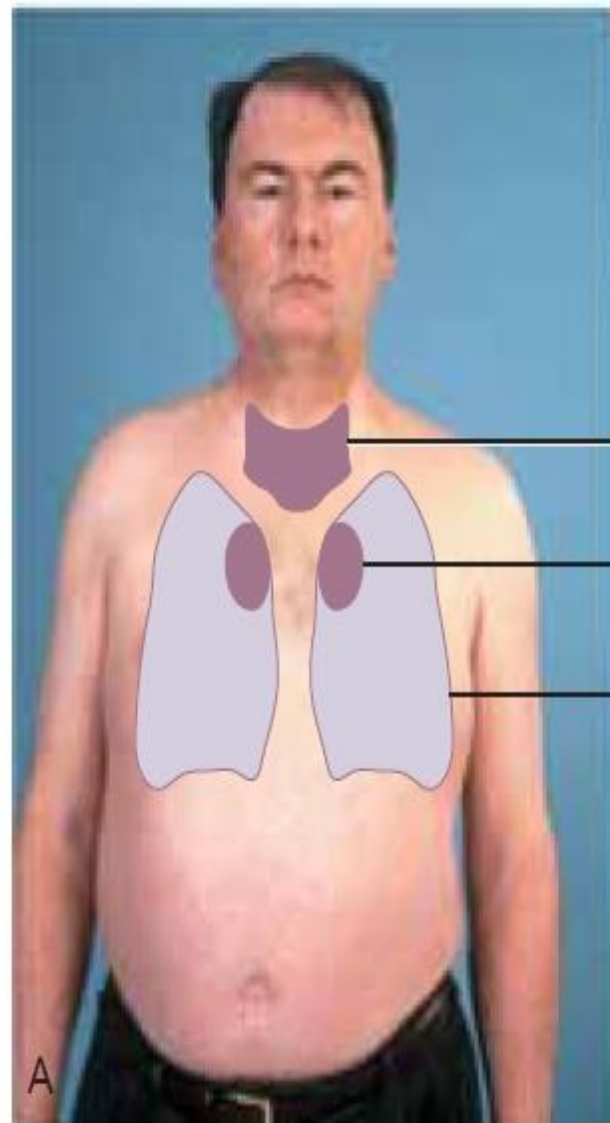
- They are normally heard over first to second interspaces anteriorly and between scapula posteriorly.
- Vesicular breath sounds are soft and low pitched, with a long inspiratory phase and a short expiratory phase.
- They are heard over most lung fields with no adventitious sounds, lungs are clear to auscultation. No crackles, wheezes, or rubs.

- Bronchial sounds heard outside of their normal locations: Fluid or consolidated tissue, such as in pneumonia.
- Diminished breath sounds: Thick, obese, or muscular chest wall, poor inspiratory effort, emphysema with air trapping, pleural effusion.
- Absent breath sounds: Missing lung/lobe, airway obstruction, pneumothorax.
- Crackles/Rales: Pulmonary edema, pneumonia, atelectasis, and upon arising in elderly people.

Auscultation

- Evaluate for normal sounds

Sound	Pitch	Intensity	Quality	I:E	Location
Bronchial	High	Loud	Blowing/ hollow	$I < E$	Trachea
Bronchovesicular	Moderate	Moderate	Combination	$I = E$	Between scapulae, 1 st & 2 nd ICS lateral to sternum
Vesicular	Low	Soft	Gentle rustling/ breezy	$I > E$	Peripheral lung

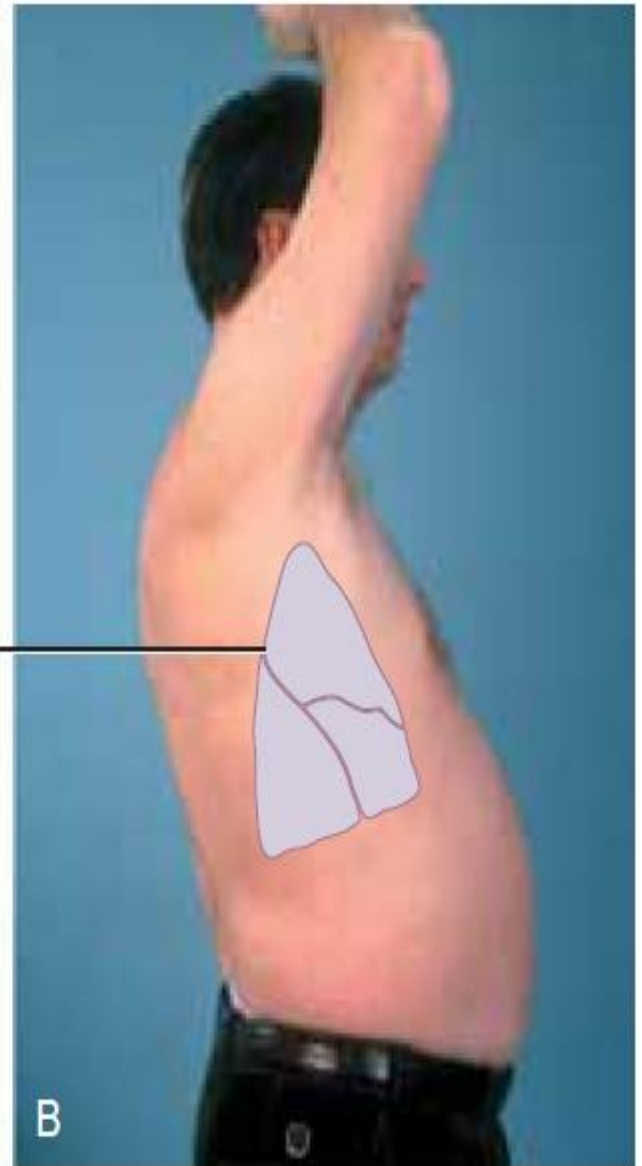


Bronchial

Broncho-
vesicular

Vesicular

Vesicular



adventitious (additional) sounds

- According to the duration adventitious sounds are classified as:
- discrete, noncontinuous sounds (crackles) and
- continuous musical sounds (wheezes).
- Pleural friction rubs are specific examples of crackles
- Friction rubs result from inflammation of the pleural surfaces
- A friction rub is best heard over the lower lateral anterior surface of the thorax.

- **Crackles (rales) are discrete, noncontinuous**
- sounds that result from delayed reopening of deflated airways.
- reflect underlying inflammation or congestion and are often present in such conditions as pneumonia, bronchitis, heart failure, bronchiectasis, and pulmonary fibrosis.

- **Wheezes are associated with bronchial wall oscillation and** changes in airway diameter.
- Wheezes are commonly heard in patients with asthma, chronic bronchitis, and bronchiectasis.

Auscultation

- Evaluate for adventitious sounds

Sound	Intensity/ Pitch	I/E	Quality	Clear with Cough
Crackles/ Rales	Soft (fine)/ High Loud (coarse)/ Low	I	Discontinuous, nonmusical, brief	Possibly
Wheeze	High	E	Continuous musical sounds	Possibly
Ronchi	Low	E	Continuous snoring sounds	Possibly
Pleural Friction Rub		I & E	Continuous or discontinuous creaking or brushing sounds	Never
Stridor		I	Continuous, crowing	Never

- [Videos of thorax and Lungs](#)

- Diagnostic modalities for respiratory system
 - Chest x-ray
 - Arterial blood gas analysis (ABGA)
 - Capnography
 - Pulse oxymetry

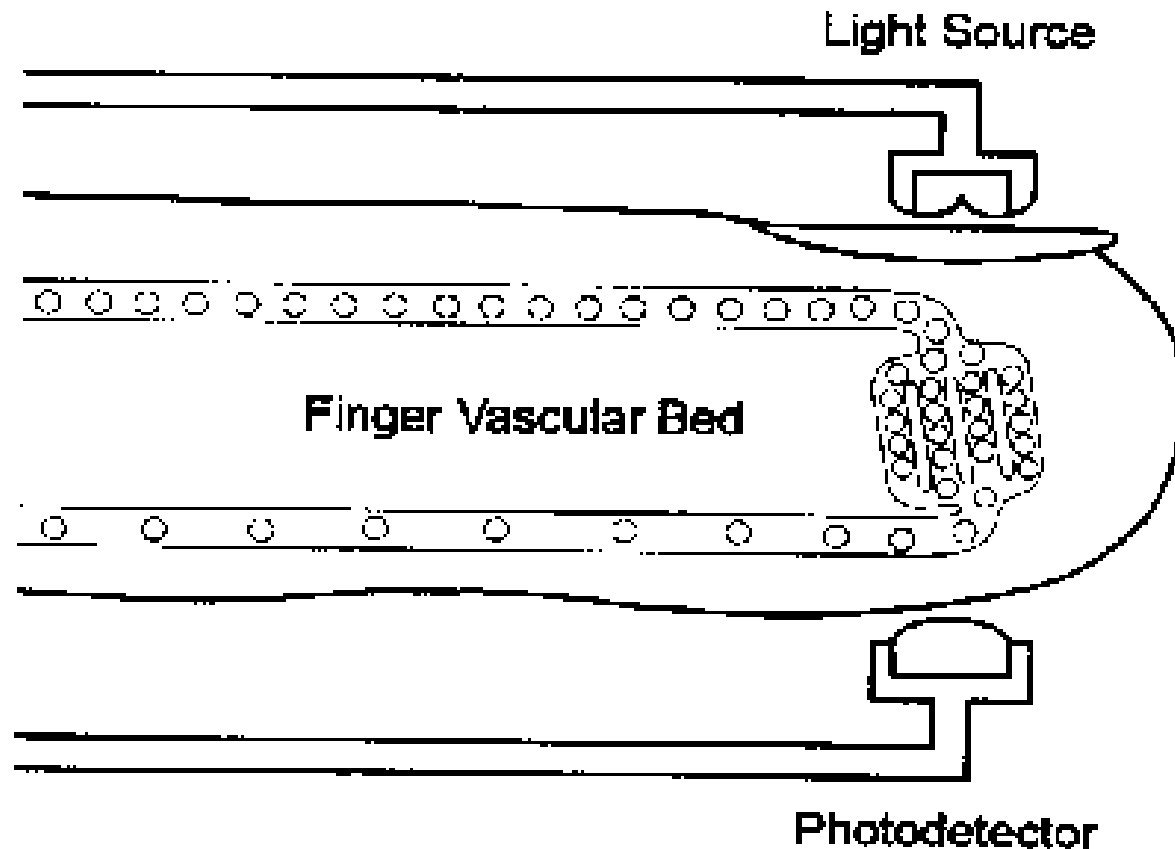
Pulse Oximetry

- Pulse oximetry is a noninvasive monitoring technique used to estimate the measurement of arterial oxygen saturation (SpO_2) of hemoglobin (also measures pulse rate).
- Oxygen saturation is an indicator of the percentage of hemoglobin saturated with oxygen at the time of the measurement .
- The reading, obtained through pulse oximetry, uses a light sensor containing two sources of light (red and infrared) that are absorbed by hemoglobin and transmitted through tissues to a photodetector.

Pulse Oximetry - Mechanics

- Light source is applied to an area of the body that is narrow enough to allow light to traverse a pulsating capillary bed and sensed by a photo detector
- Each heartbeat results in an influx of oxygen saturated blood which results in increased absorption of light
- Microprocessor calculates the amounts of HgbO₂ and reduced Hgb to give the saturation

Typical pulse oximeter sensing configuration on a finger



Probes for fingers and ear lobes are commonly used

Functional vs Fractional

- Pulse ox yields **functional saturation**
 - Ratio of HgbO₂ to the sum of all functional hemoglobins (not CO-Hgb)
 - Sites filled/sites available for O₂ to stick
- **Fractional saturation** measured by co-oximetry by blood gas analysis
 - Ratio of HgbO₂ to the sum of all hemoglobins

Saturation (SpO₂) and PaO₂

- | <u>Saturation</u> | <u>PaO₂</u> | |
|-------------------|------------------------|--------------------------------------|
| 100 | 100+ | |
| 95 | 75 | |
| <u>90</u> | <u>60</u> | |
| <u>75</u> | <u>40</u> | (mixed venous blood in pulm. artery) |
| 60 | 30 | |
| <u>50</u> | <u>27</u> | (Hb P50 point) |
- Very rough rule – PaO₂: 40,50,60 for Sat.: 70,80,90

Box 46-2

Values Affecting Pulse Oximetry

- Motion artifact
- Environmental light (e.g., sunlight, fluorescence)
- Anemia
- Deeply pigmented skin
- Carboxyhemoglobin, methemoglobin
- Nail polish
- Blood-borne dyes

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A pulse oximeter gives no information on any of these other variables:

- The oxygen content of the blood
- The amount of oxygen dissolved in the blood
- The respiratory rate or tidal volume i.e. ventilation
- The cardiac output or blood pressure

Capnography

- Used to monitor ventilation by measuring amount of expired CO_2 in the expired gas
- Two form or type of analyser
 - Mainstream and side stream
- Normal value 35-45 mmHg

monitoring of respiration by laboratory

- ABG analysis
- Invasive procedure

What is an ABG

- ✓ Arterial Blood Gas
- ✓ Drawn from artery- radial, brachial, femoral
- ✓ It is an invasive procedure.
- ✓ Caution must be taken with patient on anticoagulants.
- ✓ Arterial blood gas analysis is an essential part of diagnosing and managing the patient's oxygenation status, ventilation failure and acid base balance.

COMPONENTS OF THE ABG

- pH: Measurement of acidity or alkalinity, based on the hydrogen (H^+)-7.35 – 7.45
- Pao₂ -The partial pressure oxygen that is dissolved in arterial blood- 60-100 mm Hg.
- PCO₂: The amount of carbon dioxide dissolved in arterial blood- 35– 45 mmHg
- HCO₃⁻- The calculated value of the amount of bicarbonate in the blood 22 – 26 mmol/L
- B.E: The base excess indicates the amount of excess or insufficient level of bicarbonate. -2 to +2mEq/L-(A negative base excess indicates a base deficit in blood)
- SaO₂:The arterial oxygen saturation.->95%

Upper Respiratory System Disorders

Rhinitis

Sinusitis

Tonsillitis

Pharyngitis

Laryngitis

ACUTE RHINITIS-Rx

❖ *First line*

- Chlorpheniramine, 4mg P.O. TID for adults; 2mg P.O. 2-3 times daily for children.

❖ OR

- Cetrizine Hydrochloride, 10mg P.O. QD for adults and children above 12 years;
- 5-10mg P.O. QD for Children below 12years.
- PLUS
- Xylomethazoline, 2-3 drops of 1% solution 3-4 times a day for adults and over 6 years of age; 1-2 drops of 0.5% daily in each nostril for infants and small children up to 6 years of age.

Acute viral sinusitis

❖ Treatment

- ✓ Analgesics
- ✓ Nasal irrigation with saline solutions
- ✓ Topical corticosteroids
- ✓ Topical decongestants

Acute viral sinusitis...

❖ Topical decongestants

- Xylometazoline, 2 - 3 drops of 1% 2 - 3 times daily into each nostril. For infants and small children; 0.5% 1 - 2 drops 1 – 2 times a day into each nostril ----- PULUS

❖ Antihistamines

- Cetrizine, Adult and children above 12 years of age; 10mg P.O. QD. Children below 12 years of age; 5 – 10 mg P.O. QD--- OR
- Loratadine (Claritine):
 - Body weight >30kg - 10mg P.O. QD
 - Body weight <30kg –5mg P.O. QD

Acute viral sinusitis....

- ❖ Alternative of the above treatment
- ❖ Topical corticosteroids
 - Beclomethasone, 2–4 inhalations TID OR QID for adults; 1- 2 inhalation TID OR QID maximum of 10 inhalation for children age 6-12 years

Bacterial sinusitis

- *First line*- Amoxicillin, Adults; 500mg P.O. TID for 10 days.
- *Alternative*- Amoxicillin + Clavulanate,
 - Adults; 375mg P.O. TID for 10 days Or 625mg P.O. BID for 10 days (depending on the severity).
 - Children; 156mg /5ml or 312mg/5ml P.O. TID

Bacterial sinusitis ...

- OR- Sulfamethoxazole+trimethoprim,
- For adults-800 mg/160 mg P.O. BID for 5-7 days.
- For children 6 weeks – 5 months; 100/20 mg; 6 months – 5 yrs, 200/40 mg; 6 – 12 yrs, 400/80 mg BID.
- OR-Erythromycin, Adults; 250-500mg P.O. QID.
Children 30-50mg/kg P.O. QID

ACUTE TONSILLITIS

- ❖ Viral: Only symptomatic therapy
- ❖ Bacterial
 - *First Line*-Amoxicillin
 - *Alternative*-Ampicillin, Adults; 250-500mg P.O. QID for adults; Children 6-12 years old; 50 - 100mg/kg P.O. TID for 10 days, Children below 6 years old; 100 – 200mg/kg IV TID in divided doses.

ACUTE TONSILLITIS....

- OR- Amoxicillin + Clavulanate, Adults; 375mg P.O. TID for 10 days OR 625mg P.O. BID for 10 days. Children; 156mg/5ml P.O. BID for 10 days OR 312mg/5ml P.O. TID for 10 days
- OR-Procaïne penicillin, 800 IU IM QD for 5 – 7 days.
- OR-Benzathine Penicillin, 1,2-2,4mill Units IM once. (N.B. Known Valvular Heart Disease and rheumatic fever such treatment of Benzathine Penicillin should be continuing for prolonged period.)
- OR-Clindamycin, 150 - 450mg P.O. QID OR 300mg IM OR IV QIDOR TID for adults. 20mg/kg/day P.O. divided in 3-4doses for children.
 - In severe Infections; 20mg/kg/24hr. IM OR IV into 4 doses

ACUTE LARYNGITIS

- usually of viral origin but often followed by bacterial infection
 - Non - drug treatment: Voice rest
 - Drug treatment
- *First line*-Amoxicillin PLUS Prednisolone, 40-60mg (based on severity) P.O. QD, the dose is reduced by 5mg every 5 days.

ACUTE LARYNGITIS...

- *Alternative*-Ampicillin, Adults; 250 – 500mg P.O. QID for ten days. Children; 50 – 100mg/kg P.O. QID OR 100 – 200mg/kg IV TID
- OR- Sulfamethoxazole+trimethoprim,
- OR-Amoxicillin + Clavulanate
- OR-Clindamycin
- Plus- prednisolone

Lower Respiratory System Disorders

Bronchitis/Acute tracheo-bronchitis

Pneumonia

TB

Asthma

COPD and emphysema

Bronchiectasis

Lung abscess

Empyema

Lower Respiratory System Disorders...

Pneumothorax

pleural effusion

Atelectasis

Corpulmonale

Pulmonary embolism

Pulmonary edema

Pluerisy/Pleuritis/

ARDS (Acute Respiratory Distress Syndrome)

Acute Respiratory Failure

Lung cancer

Bronchitis/Acute tracheo-bronchitis

- ❖ Acute tracheobronchitis is an acute inflammation of the mucous membranes of the **trachea & the bronchial tree.**
- **Causative agents:** - it is often virus but the common bacterial causes are:-
 - ✓ Streptococcus pneumoniae
 - ✓ H. Influenza
 - ✓ Mycoplasma pneumonia
- Often follows URTI

Cont---

- N.B. Allergic, inhalation of chemical irritants, gases & other air contaminants can also cause bronchitis

➤ **Predisposing factors**

- ❖ Pre- existing URTI

Clinical Manifestations

- ✓ Initially patient has a **dry, irritating cough & expectorates scanty** mucoid sputum
- ✓ As the infection progresses, inspiration may become noisy (inspiratory Stridor) & cough with more **profuse purulent /thick** sputum may be present.
- ✓ Others: Fever, headache, & generalized malaise.

Medical Management

❖ For cough

- Dexamethorphan hydro-bromide 15-30mg P.O 3 to 4 times a day , syrup 15 mg/5 mL

Alternative

- Codeine phosphate 10-20 mg P.O. 3-4 times a day

❖ Antibiotic Treatment

- It is indicated, when bronchitis is complicated by bacterial infections.

Cont---

■ Antibiotics

- Amoxicillin 500gm P.O tid, for 7 days
- Erythromycin 500mg P.O Qid for 7 days
- Doxycycline 100 mg PO bid for 7 days

Nursing Interventions

- Encourage frequent coughing to remove secretions
- Advice bed rest.
- Steam inhalation
- Advice fluid intake to thin the viscous & tenacious secretions.

Pneumonia

❖ Outline

- Definition
- Causes
- Types
- Clinical manifestation
- Management

PNEUMONIA

- Is an inflammation of lower respiratory tract that involve the lung parenchyma including alveoli and supportive structure.

➤ Causes:-

- Bacteria
- Fungi
- Parasites
- viruses.

Mode of transmission

Microorganism can reach to the lung by three ways :

- 1. Aspiration**
- 2. Inhalation**
- 3. Circulatory / homogenous /**

Predisposing factors

- Immune deficiency (cancer, AIDS.)
- Congestive heart failure
- Pulmonary edema
- Age
- Malnutrition
- Obstruction of air flow

Classification

- Based on anatomical involvement
 - Lobar, broncho, interstitial, miliary pneumonia
- Based on etiology
 - Bacterial, viral, fungal, parasitc
- Based on clinical manifestation :
 - Typical, Atypical pneumonia
- Based on major occurrence,:
 - Community-acquired pneumonia
 - Hospital acquired pneumonia,

1. Community acquired pneumonia

- Infection of the lung parenchyma in a person who is not hospitalized or living in a long-term care facility for ≥ 2 weeks.

cause;- mostly bacteria

- *S. pneumonia*, *H. influenza*, *Legionella*
- Most common pathogen = *S. pneumo* (60-70% of CAP cases)

Hospital-Acquired Pneumonia

- is occurs with in 48 hours or more after admission to the hospital.
- Result from exposure to potentially infectious agent .
- Organism that are responsible for hospital - acquired pneumonia are different from CAP. 90% of organism are bacteria.
- Hospital-acquired pneumonia (HAP)
 - Occurs 48 hours or more after admission, which was not incubating at the time of admission
- Ventilator-associated pneumonia (VAP)
 - Arises more than 48-72 hours after endotracheal intubation

- Healthcare-associated pneumonia (HCAP)
 - Patients who were hospitalized in an acute care hospital for two or more days within 90 days of the infection; resided in a nursing home or LTC facility; received recent IV abx, chemotherapy, or wound care within the past 30 days of the current infection; or attended a hospital or hemodialysis clinic

The common responsible Pathogens

- *P. aeruginosa*,
- *Staphylococcus aureus*.
- *Enterobacter* species
- *Escherichia coli*
- *Klebsiella* species

- **Pathophysiology of lobar pneumonia**

- ❖ **Edema phase/Congestion** – inflammatory process

- ↑ fluid produced by alveoli ,bronchiole and bronchi,
- Bacteria grow & spread and filled with exudates that prevent ventilation and diffusion.

- ❖ **Red hepatization phase**

- ✓ Erythrocytes in the cellular intra-alveolar exudate
- ✓ Neutrophil influx of host defense
- ✓ Bacteria are occasionally seen
- ✓ Lung appears red like liver.

❖ Gray hepatization phase

- ✓ No new erythrocytes are extravasating , and lysed and degraded. The neutrophil is the predominant cell, fibrin deposition is abundant, and bacteria have disappeared

❖ Resolution phase

- ✓ The macrophage reappears as the dominant cell type in the alveolar space
- ✓ The debris of neutrophils, bacteria, and fibrin has been cleared

Lobar pneumonia Con't

2. Red Hepatization –

3. Gray Hepatization –



Typical pneumonia

Have clinical presentations. Like:

- Sudden onset of shaking chills
- Cough (non productive / productive)
- Tachypnea 25-45b/m, use of accessory muscle (dyspnea).
- Rapidly rising fever (38.5° to 40.5°C)
- Pleuritic chest pain
- WBC count is elevated
- Chest x-ray – opacity

Atypical pneumonia

- Pneumonia with atypical clinical radiologic manifestations and is due to mycoplasmas ,ricketisia ,PCP ,etc
- Upper respiratory tract infection that the onset of symptoms is gradual and non specific.

The predominant symptoms:

- Headache
- low-grade fever
- pleurisy chest pain
- Myalgia, muscle soreness and fatigue
- Dry cough with some mucoid sputum

Clinical presentation

- **Symptoms & Signs**
 - Most typical signs/symptoms
 - ✓ Fever
 - ✓ Cough (non productive / productive)
 - ✓ Pleuritic chest pain
 - ✓ Chills and/or rigors
 - ✓ Dyspnea

– Frequent signs/symptoms

- Headache
- Nausea
- Vomiting
- Diarrhea
- Fatigue

- ✓ Arthralgia/myalgia
- ✓ Falls and new-onset or worsening confusion

- **Physical findings**

- ✓ Fever

- ✓ Tachypnea

- ✓ Tachycardia

- Chest examination

- ✓ Dullness to percussion

- ✓ Increased tactile and vocal fremitus

- ✓ Egophony

- ✓ Crackles

- ✓ Pleural friction rub

Diagnostic Findings

- Hx of recent respiratory tract infection
- physical examination(crackles)
- chest x-ray studies (consolidation), opacity
- blood culture, neutrophils increased
- sputum exam

Indication for admission

There are 6 criteria to be admitted.

1. Marked V/S abnormality
(HR>140,BP<90,RR>30)
2. Altered mental status
3. Partial oxygen concentration < 60mmHg
4. Complicated pneumonia
5. Sever acute metabolic abnormality
6. Acute co-existing medical conditio.

Indication for admission

- Another method to admit patient is based on:
 - Severity of illness score
 - CURB-65: confusion, uremia, respiratory rate, low BP, age 65 or greater
 - if >2 consider hospitalization

Management

Recommended for treatment of outpatients with no heart problem include:

Antibiotics

- Pencillin G, crystalin pencillin, pr.pencillin.
- Cephalosporin groups e.g. ceftraxone
- Amoxicilin
- Erythromycin
- Doxycycline
- Co-otrmoxazole for PCP

Nursing care

- Check potency of drugs
- Check oxygenation
- Provide oxygen if hypoxia.
- Check fluid intake
- Perform postural drainage to mobilize secretion
- Assess vital sign every 4 hours.
- Provide coughing and berating exercise.

TUBERCULOSIS

Learning objectives

- ✓ Define Tuberculosis
- ✓ List the causative agent of TB
- ✓ Explain the different types of TB
- ✓ Explain the mode of transmission of TB
- ✓ Identify the clinical features of TB
- ✓ Discuss the methods of diagnosis of TB
- ✓ Explain the treatment of TB
- ✓ Discuss the prevention method of TB

Definition

- ✓ Tuberculosis (TB) is an air borne infectious disease caused by any one of a group of bacteria collectively known as the *Mycobacterium tuberculosis* complex.
- ✓ These complexes are
 - (*M. Tuberculosis*, *M. bovis*, *M. africanum*, *M. microti*, *M. caprae*, *M. pinnipedii*, *M. canetti* and *M. mungi*)
- ✓ **Most of the cases, the disease is caused by *M. Tuberculosis***

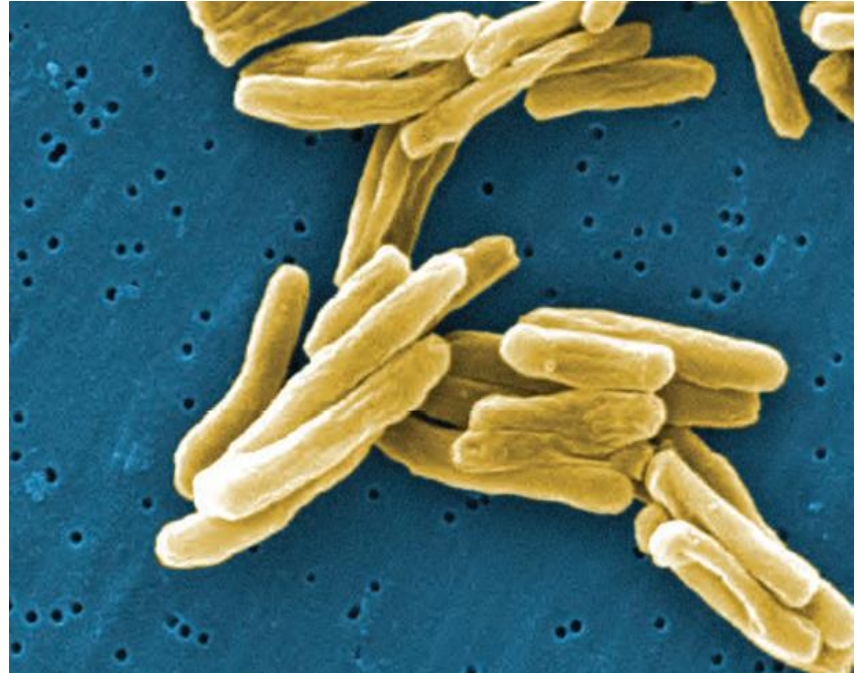
TB definition.....

- ✓ Occasionally the disease can also be caused by *Mycobacterium bovis* and *Mycobacterium africanum*
- ✓ *M. tuberculosis* is usually affecting the lungs in which case it is called pulmonary TB.
- ✓ Mainly transmitted person-to-person by inhalation of infected **droplet nuclei**

Characteristics of TB Bacteria

➤ TB bacteria are:

- Aerobic
- Rod-shaped
- 1-5 microns in size
- Aerobic
- Cell wall high lipid content (needs specific laboratory methods – Acid fast bacilli smear and Mycobacterium culture)



Epidemiology

- TB affects individuals of all ages and both sexes, and estimated to infect one-third of world population
- Usually affects economically and culturally disadvantaged segment of a population where access to health services is often limited.
- Common among HIV patients
- It is still the single most infectious agent that takes more life
- Emergence of drug resistant TB (MD and XDR-TB)

Mode of transmission

- *M. tuberculosis* is carried in airborne particles, called droplet nuclei.
- The nuclei are generated when persons who have pulmonary or laryngeal TB disease cough, sneeze, shout, or sing.
- Depending on the environment, these tiny particles can remain suspended in the air for several hours.

Mode of transmission...

- *M. tuberculosis* is transmitted through the air, **not** by surface contact.
- Transmission occurs when a person inhales droplet nuclei containing *M. tuberculosis*, and the droplet nuclei traverse the mouth or nasal passages, upper respiratory tract, and bronchi to reach the alveoli of the lungs
- Consumption of raw milk containing *M. bovis*

Risk factors of TB

- The risk of acquiring of TB infection is determined mainly by **exogenous factors**
- The risk of developing Of TB disease after being infected depends largely on **endogenous factors**

- ✓ Most infectious patients have cavitary pulmonary disease
- ✓ Up to 20 contacts may be infected by each AFB-positive case before the index case is found to have tuberculosis
- ✓ The risk of acquiring *M. tuberculosis* infection is determined mainly by **exogenous factors**

From infection to disease

- The risk of developing disease after being infected depends largely on **endogenous factors**
- Individual's innate immunity defences and level of function of cell-mediated immunity (CMI)
- Age is an important determinant of the risk of disease after infection
- ~10% of infected persons will eventually develop active tuberculosis in their lifetime

Factors that Determine the Probability of Transmission of M. tuberculosis

Factor	Description
Susceptibility	• Susceptibility (immune status) of the exposed individual
Infectiousness	• Infectiousness of the person with TB disease is directly related to the number of tubercle bacilli that he or she expels into the air. • Persons who expel many tubercle bacilli are more infectious than patients who expel few or no bacilli
Environment	• Environmental factors that affect the concentration of M. tuberculosis organisms
Exposure	• Proximity, frequency, and duration of exposure

Characteristics of a Patient with TB Disease that Are Associated with Infectiousness

Factor	Description
Clinical	• Presence of cough, especially lasting 3 weeks or longer • Respiratory tract disease, especially with involvement of the larynx (highly infectious) • Failure to cover the mouth and nose when coughing • Inappropriate or inadequate treatment (drugs, duration)
Procedure	• Undergoing cough-inducing or aerosol-generating procedures (e.g., bronchoscopy, sputum induction, administration of aerosolized medications)
Radiographic and laboratory	• Cavitation on chest radiograph • Positive culture for <i>M. tuberculosis</i> • Positive AFB sputum smear result

Env'tal Factor	Description
Concentration of infectious droplet nuclei	The more droplet nuclei in the air, the more probable that M. tuberculosis will be transmitted (Talking 0 – 200, Coughing 0 – 3 500, Sneezing 4500 – 1, 000, 000)
Space	Exposure in small, enclosed spaces
Ventilation	Inadequate local or general ventilation that results in insufficient dilution or removal of infectious droplet nuclei
Air circulation	Recirculation of air containing infectious droplet nuclei
Specimen handling	Improper specimen handling procedures that generate infectious droplet nuclei
Air Pressure	Positive air pressure in infectious patient's room that causes M. tuberculosis organisms to flow to other areas

Proximity and Length of Exposure Factors

Factor	Description
Duration of exposure to a person with infectious TB	• The longer the duration of exposure, the higher the risk for transmission
Frequency of exposure to infectious person	• The more frequent the exposure, the higher the risk for transmission
Physical proximity to infectious person	• The closer the proximity, the higher the risk for transmission

Natural history- in untreated tuberculosis

- About one-third of patients die within 1 year after diagnosis, and more than 50% died within 5 years
- Of the survivors at 5 years, ~60% had undergone spontaneous remission, while the remainder were still excreting tubercle bacilli

Natural history

Exposure to *M. tuberculosis*

Intimacy & duration of contact, the degree of infectiousness

M. tuberculosis Infection

Weakened Immunity

1-2yrs

Latent Tuberculosis

~10%

Tuberculosis Disease (Case)

Of survivors at 5yrs

60%

Spontaneously Cured

40%

Complicated and still excreting bacilli

1/3 in 1yr
1/2 in 5 yrs

Death

Pathogenesis

- Inhalation and deposition in the lungs of the tubercle bacillus leads to one of **four possible** outcomes:
 - Immediate clearance of the organism
 - Chronic or latent infection
 - Rapidly progressive disease (or primary disease)
 - Active disease many years after the infection (reactivation disease)

- ✓ The tubercle bacilli establish infection in the lung after they are carried in droplets to reach the alveolar space
- ✓ If the **innate defence** system of the host fails to eliminate this infection
- ✓ The **bacilli proliferate** inside alveolar macrophages and kill the cells

- ✓ The infected macrophages produce **cytokines and chemokines** that attract other phagocytic cells
- ✓ Including monocytes, other alveolar macrophages, and neutrophils, which eventually form a **nodular granulomatous structure called the tubercle**

- ✓ If the bacterial replication is not controlled, the **tubercle enlarges** and the bacilli enter the local draining lymph nodes
- ✓ This leads to lymphadenopathy, a characteristic manifestation of primary TB
- ✓ The lesion produced by the expansion of the tubercle into the **lung parenchyma** and **lymph node** involvement is called the **Ghon complex**

- ✓ The bacilli continue to proliferate until an effective cell-mediated immune (CMI) response develops, usually **two to six weeks** after infection
- ✓ Failure by the host to mount an effective CMI response and tissue repair leads to **progressive destruction of the lung**
- ✓ Effective CMI and delayed type hypersensitivity reaction leads to inhibition of tubercle bacilli, *caseous necrosis* and healing, which later may calcify
- ✓ Tubercle bacilli may survive inside the **healed lesion**

- **Classification of TB**

- ✓ Is broadly classified in to:

- **PTB** :accounts 85% of all Tb cases

- ✓ Smear positive PTB (75 – 80 %)

- ✓ Smear negative PTB (20 – 25 %)

- **EPTB**: accounts 14 % of all Tb cases

Clinical manifestation

Pulmonary TB:

- ✓ A person is a suspect of Pulmonary Tuberculosis when presenting with persistent cough for two weeks or more
- ✓ Cough is usually with expectoration, with or without blood stained sputum and can be accompanied by one or more of the following symptoms:

- ✓ Weight loss
- ✓ Chest pain
- ✓ Shortness of breath
- ✓ Intermittent fever
- ✓ Night sweats
- ✓ Loss of appetite
- ✓ Fatigue and malaise

- ✓ Pulmonary tuberculosis-can be categorized as **primary or postprimary** (secondary)
- ✓ Primary PTB occurs soon after the initial infection with tubercle bacilli
- ✓ In children and in persons with impaired immunity, primary pulmonary tuberculosis may progress rapidly to clinical illness

Postprimary /reactivation/ adult type TB

- Results from endogenous reactivation of latent infection and is usually localized to the apical and posterior segments of the upper lobes
- The extent of lung parenchymal involvement varies greatly, from small infiltrates to extensive cavitory disease

Extra-pulmonary TB

- The signs and symptoms of *Extrapulmonary Tuberculosis (EPTB)* depend mainly on the organ(s) involved.

Extra PTB site

- ✓ lymph nodes - 35%
- ✓ Pleural TB - 20 %
- ✓ Genitourinary tract - 10 –15 %
- ✓ Bones and joints - 10 %
- ✓ Meninges - 5 %
- ✓ Gastrointestinal TB - 3.5 %
- ✓ Pericardial TB - 1 – 2 %

Diagnosis methods of tuberculosis

- **Smear Microscopy**

- ❖ Two staining methods can be used to identify AFB
- ✓ Ziel-Neelsen staining (ZN) : has low sensitivity (40-60%) and requires 5,000-10,000 bacilli per ml of sputum to get positive results.
- ✓ Fluorescence auramine staining (LED FM): requires less time for slide reading and has additional 10% sensitivity over ZN microscopy to identify bacillus

- ✓ Culture and drug sensitivity testing- for definite diagnosis , takes 6-8 weeks
- ✓ Chest x-ray- upper lobe infiltrates, bilateral infiltrates, cavitation, pulmonary fibrosis, interstitial infiltrates, intrathoracic LAP
- ✓ Skin testing (TST)
- ✓ PCR
- ✓ Interferon- γ release assay- in low incidence
- ✓ Adenosine deaminase test (ADA)

- **Drug susceptibility testing (DST):** is required to make a definitive diagnosis of drug resistant TB.
- It can be done either by phenotypic or genotypic techniques

Treatment of TB

- ❖ The main objectives of anti-TB treatment are to:
 - ✓ Cure the patient of TB (by rapidly eliminating most of the bacilli)
 - ✓ Prevent death from active TB or its late effects
 - ✓ Prevent relapse of TB (by eliminating the dormant bacilli)
 - ✓ Prevent the development of drug resistance (by using a combination of drugs)
 - ✓ Decrease TB transmission to others.

Standard First Line regimens

Table 7. Showing the Standard TB treatment regimen in Ethiopia

TB Patient type	Standard Regimen		Patient registration groups receiving the regimen
	Intensive phase	Continuation phase	
New TB case	2(RHZE)	4(RH)	• New TB patients
	2(RHZE)S	10 (RH)	• New patients with CNS TB(meningitis, tuberculoma)
	2(RHZE)	10 (RH)	• New TB patients involving vertebra and Osteoarticular space
Previously treated TB case	2(RHZE)S, (RHZE)	5(RHE)	• Relapse • Treatment after LTFU • Treatment after failure of New regimen • Others

Table 6. Showing First line Anti-TB Drug formulations for Adult & those weighing >25kg

DRUGS	FORMULATION	STRENGTH(mg)	Preparation, route
HRZE	Tablet	75/150/400/275	FDC, oral
HR	Tablet	75/150	FDC, oral
HRE	Table	75/150/275	FDC, oral
E	Tablet	400	Loose, oral
STM	Powder for injection	1000	Loose, Parental

Table 7. Showing the Standard first line Anti-TB treatment regimen in Ethiopia

TB Patient type	Standard Regimen		Patient registration groups receiving the regimen
	Intensive phase	Continuation phase	
Drug susceptible TB case (New and Previously treated)	2(RHZE)	4(RH)	• New TB patients • Relapse • Treatment after LTFU • Treatment after failure of New regimen • Others
	2(RHZE)	10 (RH)	• New patients with CNS TB(meningitis, tuberculoma) • New TB patients involving vertebra and Osteoarticular space

* previously treated TB patients are treated with the same regimen as for New TB patients for six months after ruling out the presence of resistance at least for Rifampicin with rapid DST.

Phases of chemotherapy

- **Intensive (initial) phase**
 - It renders the patient non-infectious by rapidly reducing the load of bacilli , usually within two weeks
 - The regimen in this phase consists of:
 - ✓ Combination of four drugs for the first eight weeks for new cases, or
 - ✓ Combination of five drugs for the first eight weeks followed by four drugs for the next four weeks for previously treated cases.

- ***Continuation phase***

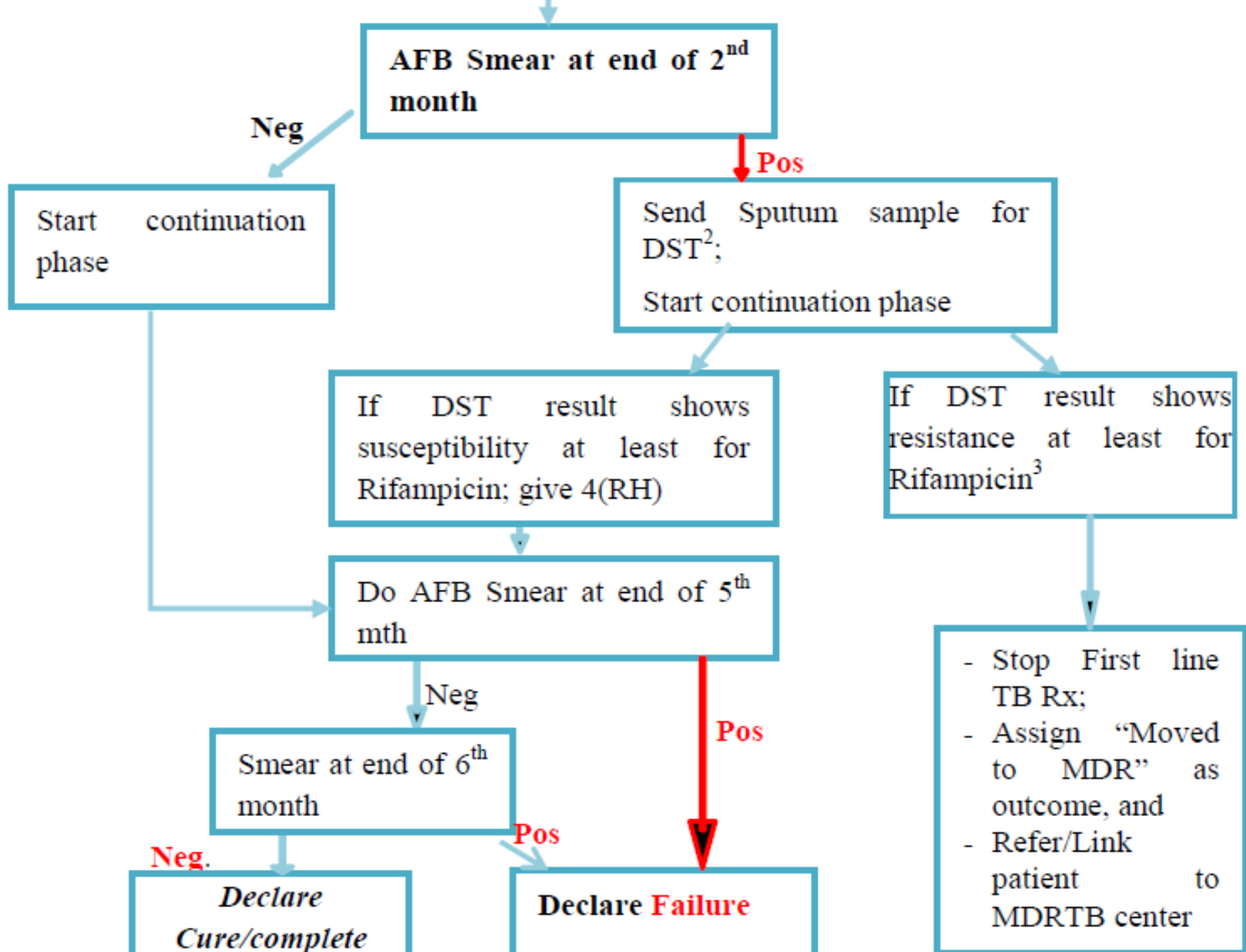
- ✓ This phase aims to sterilize the remaining semi-dormant bacilli and is important to ensure cure/ completion of treatment and prevent relapse after completion of treatment.
- The regimen in this phase consists of:
 - Combination of two drugs to be taken for 4 months for new cases, or
 - Combination of three drugs to be administered for 5 months for previously treated cases.

Dosage of new regimen for New adult TB cases: 2(RHZE)/ 4(RH)

Patients weight band (Kg)	Treatment regimen and Dose	
	Intensive phase: 2(RHZE)	Continuation Phase: 4(RH)
20-29	1 ½	1 ½
30-39	2	2
40-54	3	3
≥55	4	4

Dosage of retreatment regimen for adult TB cases: 2S (RHZE),1(RHZE) / 5(RH)E

Patients Weight band (Kg)	Treatment regimen and dose			
	Intensive phase :		Continuation phase:	
	2S(RHZE),(RHZE)		5(RH)E	
	(RHZE)	S	(RH)	E
20-29	1 ½	500mg	1 ½	1 ½
30-39	2	500mg	2	1 ½
40-54	3	750mg	3	2
≥ 55	4	1000mg	4	3



Follow up, New PTB patient

- **At the end of 2 month**

- ✓ If smear-positive at month 2, obtain sputum again at month 3
- ✓ If smear-positive at month 3, obtain culture and DST.

- **At the end of 5 month**

- ✓ if sm+, obtain culture and DST

- **At the end of 6 month**

- ✓ if sm +, obtain culture and DST

Follow up, Retreatment patient

- At the end of 3 months
 - ✓ if sm +, obtain culture and DST
- At the end of 5 months
 - ✓ if sm +, obtain culture and DST
- At the end of 8 months
 - ✓ if sm +, obtain culture and DST

Overview of MDR-TB

- **Mono Resistance:** Resistance to single drug
- **Poly-Resistance:** Resistance to more than one drug other than Rifampicin and Isoniazid together
- **Multi-Drug Resistance (MDR):** Resistance to at least Isoniazid and Rifampicin
- **Extensive Drug Resistance (XDR-TB):** MDR plus Resistance to:
 - A fluoroquinolone : Ciprofloxacin, Ofloxacin, Levofloxacin, Moxifloxacin, Gatifloxacin, etc. and
 - One or more of the following injectable agents: Kanamycin, Amikacin, capreomycin, Viomycin

Drug-Resistant TB: Definitions

- **Primary drug-resistance: “New Cases”**
 - ✓ Drug resistance in a patient who has never been treated for tuberculosis or received less than one month of therapy
- **Secondary (acquired) drug-resistance: “Previously Treated Cases”**
 - ✓ Drug resistance in a patient who has received at least one month of anti-TB therapy

Factors that Lead to Drug Resistance

Causes of Inadequate treatment:

- Patient-related factors
- Healthcare provider-related factors
- Healthcare system-related factors

Treatment of MDR-TB

Group 1: First-line	Second-line			Third-line
Isoniazid	Group 2: Injectables	Group 3: Fluoroquinolones	Group 4: Oral bacteriostatic SLDs	Group 5: Agents with unclear role
Rifampin	Streptomycin			
Ethambutol	Kanamycin	Ofloxacin Levofloxacin Moxifloxacin Gatifloxacin	Ethionamide Cycloserine PAS Prothionamide Teridizone	AMX/CLV Clofazimine High-dose INH Clarithromycin
Pyrazinamide	Amikacin			
	Capreomycin			
	Viomycin			

10.4.3 Treatment regimen and Phases of treatment

9-month-standardized shorter DR-TB treatment regimen:

4 Km-Mfx-Pto-Cfz-Z-HH-E / 5 Mfx-Cfz-Z-E

Intensive phase may be prolonged up to six months, if the patient remains smear positive after month four of treatment.

Duration of Rx

- **Intensive phase:** the IP aims for rapid clearance of the bacilli with co-administration of 7 drugs and render the patient sputum AFB smear converted preferably within the first 4 months of treatment.
- In case of non-conversion on sputum AFB microscopy at the end of month 4 of treatment, the duration of the IP may be extended by 1 month.
- The maximum duration of the IP is 6 months.

Duration of Rx

- **Continuation phase:** the CP aims to sterilize the body from remaining bacilli.
- The regimen consists of Moxifloxacin, Clofazimine, ethambutol, and pyrazinamide given daily for 5 months.
- The decision to transition from the IP to the CP requires information of at least monthly follow up smear microscopy results and the clinical responses of the patient, while awaiting sputum culture follow-up results.

BRONCHIAL ASTHMA

Learning Objective

At the end of this unit the student will be able to

- ✓ Define bronchial asthma
- ✓ Describe the etiology of bronchial asthma
- ✓ Explain the epidemiology of bronchial asthma
- ✓ Explain pathophysiology of bronchial asthma
- ✓ Identify the clinical manifestations of bronchial asthma
- ✓ List the signs of severity of bronchial asthma
- ✓ Make an accurate diagnosis of bronchial asthma
- ✓ Manage most cases of bronchial asthma
- ✓ Differentiate complicated cases of bronchial asthma

Definition:

- ✓ Is a chronic reversible air way obstruction characterized by: cough, shortness of breath, chest tightness, wheezing & rapid respirations
- ✓ It is associated with widespread airway obstruction that is reversible either spontaneously or with treatment

Etiology Asthma

- ❖ Can be classified in to 3 types
 - ✓ Allergic (atopic)
 - ✓ Non allergic (idiosyncratic)
 - ✓ Mixed type

- **Epidemiology:**

- ✓ The prevalence of asthma is rising in different parts of the world.
- ✓ It can occur **at any age**; but it usually starts **early in life**
- ✓ Asthma is diagnosed before **the age of seven years** in approximately **75 percent of cases**
- ✓ **Males are affected twice** as common as females in early life; this sex difference equalizes by age 30
- ✓ Most cases of asthma are associated with personal or family history of allergic disease such as **eczema, rhinitis and urticaria**

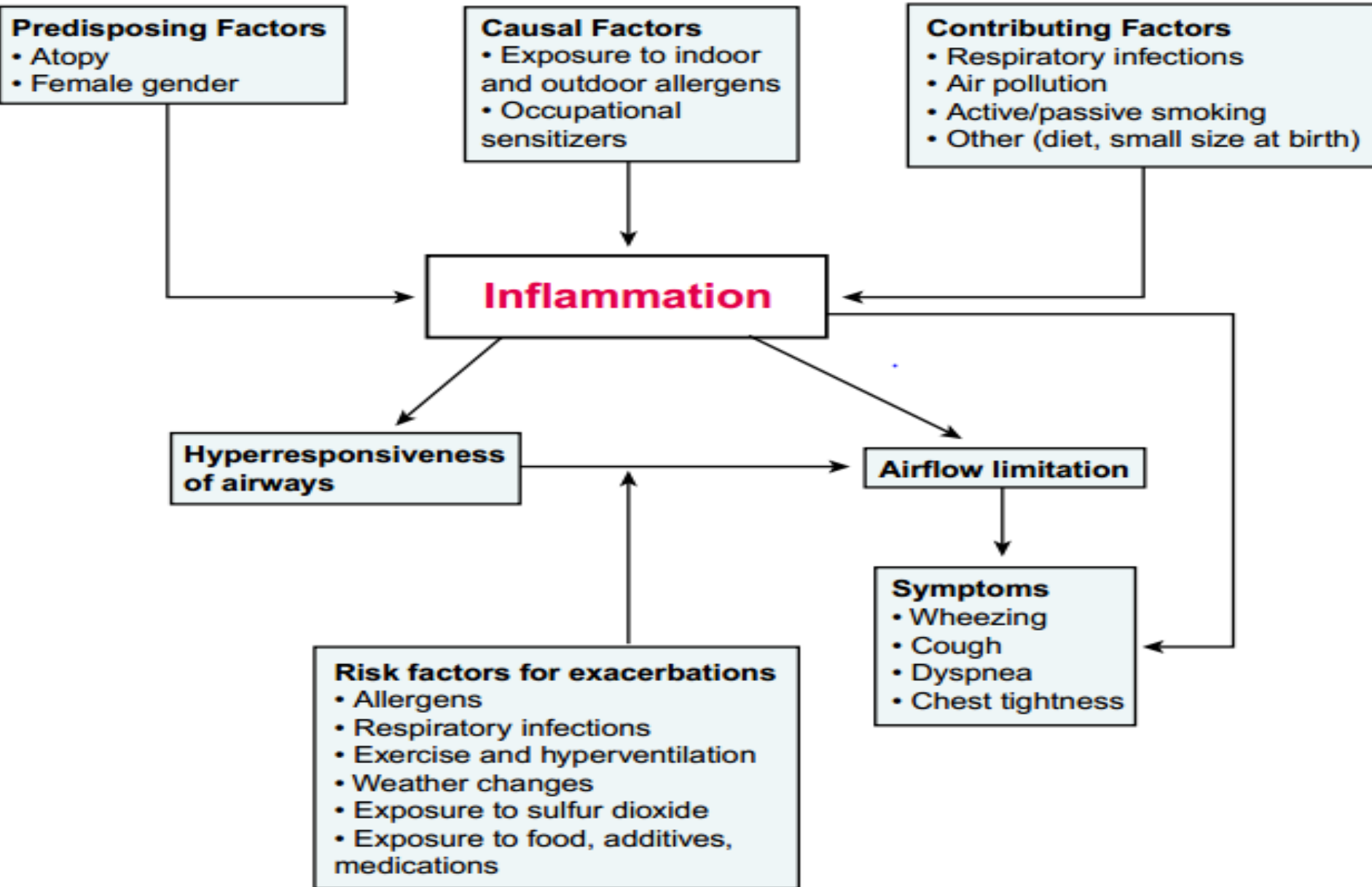
Pathophysiology

- ✓ Asthma results from a state of **persistent subacute inflammation of the airways**
- ✓ The airways obstruction in asthma is due to a **combination of factors**
- ✓ The **cells thought to play** important part in the inflammatory response are **mast cells, eosinophils, lymphocytes and airway epithelial cells**

- These cells release inflammatory mediators which may result
- ✓ Bronchoconstriction (spasm of airways smooth muscles)
- ✓ Vascular congestion and edema of airways mucosa
- ✓ Increased mucus production
- ✓ Injury and desquamation of the airways epithelium and **impaired muco-ciliary transport**



Physiology/Pathophysiology



- ✓ Cells that play a key role in the inflammation of asthma are mast cells, neutrophils, eosinophils, and lymphocytes.
- ✓ Mast cells, when activated, release several chemicals called mediators. These chemicals, which include histamine, bradykinin, prostaglandins, and leukotrienes
- ✓ Regulation of these chemicals is the aim of much of the current research regarding pharmacologic therapy for asthma.
- ✓ Alpha- and beta2-adrenergic receptors of the sympathetic nervous system are located in the bronchi.

- ✓ The **alpha adrenergic receptors** are stimulated, **bronchoconstriction** occurs; when the **beta2-adrenergic receptors** are stimulated, **bronchodilation** results
- ✓ The balance between alpha and beta2 receptors is controlled primarily by cyclic adenosine monophosphate (cAMP).
- ✓ **Alpha-adrenergic** receptor stimulation results in a decrease in cAMP, which leads to an increase of chemical mediators released by the mast cells and cause bronchoconstriction.
- ✓ **Beta 2-receptor** stimulation results in increased levels of cAMP, which inhibits the release of chemical mediators and causes bronchodilation

- Symptom and Signs

❖ **Classic triad of symptoms**

- ✓ **Wheeze** (high-pitched whistling sound, usually upon exhalation)
- ✓ **Cough** (typically worsening at night)
- ✓ **Shortness of breath** or difficulty breathing

On physical examination

- ✓ Varying degrees of respiratory **distress tachypnea, tachycardia, and audible wheezes** are often present
- ✓ **Dehydration** may be present because of sweating and tachypnea
- ✓ Chest examination shows a **prolonged expiratory** phase with relatively high pitched wheezes throughout inspiration and most of expiration
- ✓ In more severe episodes, patients may be **unable to speak** more than a few words without stopping for breath
- ✓ **Cyanosis** is usually a late sign of hypoxia
- ✓ **Confusion and lethargy** may indicate the onset of progressive respiratory failure

- Signs of severity of asthmatic attack
 - ✓ Tachycardia HR >120 /min, Tachypnea RR.30 /min
 - ✓ Presence of pulsus paradoxus
 - ✓ Use of accessory muscles of respiration
 - ✓ Cyanosis
 - ✓ Altered state of consciousness (confusion , drowsiness)

- ✓ Silent chest
- ✓ Paradoxical movement of the chest and the abdomen
- ✓ Presence of complications : Pneumothorax , atelectasis
- ✓ Unable to finish a sentence with single breath (frequent interruption of speech to take a breath)
- ✓ Laboratory parameters : $\text{SaO}_2 < 90 \%$, $\text{PaCO}_2 > 42 \text{ mmHg}$

Diagnosis method

- ✓ Diagnosis Asthma should be considered in anyone who wheezes.
- ✓ A family history of allergy, rhinitis or asthma can be elicited in most asthmatics
- ✓ Eosinophilia is a common finding.
- ✓ Chest x-ray: varies from normal to hyperinflation

MANAGEMENT OF BRONCHIAL ASTHMA

Prevention of attacks

- The role of environmental factors (e.g. animal dander, dust, airborne moulds, and pollens) in acute exacerbations is clear. Allergens that can be controlled by avoidance should be eliminated.
- Nonspecific exacerbating factors (e.g. cigarette smoke, odors, irritant fumes, and change in temperature, and humidity) should also be investigated and avoided if possible.

The Drugs for asthma therapy include

- **Bronchodilators** (adrenergic agonist, methylxanthine drugs)

❖ Salbutamol

- It is similar to terbutalin (β_2 selective)
- After inhalation significant effect is seen within 15 minutes and lasts for 3-4hrs

Dosage- oral- adults 2-4mg 3-4 times daily

Children-2-6yrs 1-2mg 3 times daily

Older children 2mg 2-4 times daily]

-Inhalation- 1-2 inhalation (100mg dose) Q4hrs

-Parental- 500mg sc/Im repeated every 4 hrs as required

- Slowly IV 250mg in solution containing 50mg/ml

5. TERBUTALINE SULPHATE

- It is greater selectivity for β_2 (no cardiac stimulation) and is long acting

Dosage: - orally- adults- 5mg 2-3 times daily

- Children- 1.5-3mg (0.05mg/kg/dose) 3 times daily

- SC- 0.25 mg.

❖ Theophylline

- Relaxes bronchial smooth muscle, causing **bronchodilation**
- **MOA**
 - ✓ Inhibition of **PDE**, which increases the concentration of cAMP
 - ✓ It also inhibits the release of slow reacting substances of anaphylaxis and histamine

ii. **AMINOPHILLINE** (Theophylline + Ethylenediamine)

MOA – is the same as theophylline

Indication – as that of theophylline

CI - hypersensitivity

- PUD or active gastritis

Use with **caution** – in cardiac arrhythmia, CHF

- sever HPN

- Hyperthyroidism, alcoholism, labor, lactation

Preparation – Tab

- inj.250mg/10ml

- suppository

Dosage – Adult 250-500mg orally or by slow Iv injection 10-15ml

- Children < 6 months- is not recommended

- **Anti-inflammatory - Corticosteroids**

- ✓ Are not bronchodilators and their major uses in asthma is **reducing bronchial inflammation**

MOA – Act primarily by suppressing inflammation

- a) ↓ Synthesis & release of inflammatory mediators
- b) ↓ Infiltration & activity of inflammatory cells
- c) ↓ Edema of air way mucosa (2⁰ to decrease in vascular permeability)

- **Inhalational therapy**- considered as first line therapy

- Should be used with β_2 -agonist
- Examples:- Beclomethasone 2 puffs qid

2. **Oral therapy**: - for pts with severe asthma

- **Duration** should be as **short** as possible to decrease toxicity

Drugs:- **prednisolone** 30-60 mg po/d

- **Dexamethasone**

S/E: - Use for more than 15 days can be hazardous

- Osteoporosis,
- Hyperglycemia
- PUD,
- Adrenal suppression, growth suppression

3. **Injectables** :- Hydrocortisone **3 mg/kg** iv infusion 6 hourly

- ✓ Leukotrienes inhibitors
- ✓ Inhibitors of mediator release (Mast cell stabilizers)

Complications

- ✓ Pneumothorax
- ✓ Emphysema
- ✓ Atelectasis due to obstruction
- ✓ Dilated right heart chambers (Corpulmonale) :
from chronic hypoxemia and pulmonary
hypertension
- ✓ Respiratory failure

Pleurisy/Pleuritis

- ✓ It is an inflammation of both layers of the pleura

C/M

- ✓ Severe, sharp, pleuritic chest “knifelike” pain during inspiration

➤ Friction rub

➤ The pain may be **localized or radiates** to the shoulder / abdomen

- **Pleurisy may develop with: -**
 - ✓ Pneumonia, URTI, Tuberculosis
 - ✓ Pulmonary embolism

Dx:-

- ✓ Chest X-rays
- ✓ Sputum examination
- ✓ Pleural fluid analysis / pleural biopsy

- **Medical Mx:-**

- ✓ Treat the underlying disease /pneumonia. . . /
- ✓ Analgesics / Indometacin 25mg p.o tid for 1week

- **Nursing interventions**

- ✓ Turn frequently on the affected side to splint the chest wall / this lessen the stretching of the pleura/
- ✓ Emotional support & teaching

Pleural Effusion

Learning Objective

At the end of this session you will be able to

- ✓ Define pleural effusions
- ✓ Classify pleural effusions
- ✓ List the etiologies of pleural effusions
- ✓ Identify the clinical manifestations of pleural effusion
- ✓ Describe the most commonly used method of diagnosis of pleural effusions
- ✓ Manage pleural effusions

Definition

- ✓ Pleural effusion is the presence of **excess fluid** in the pleural space
- ✓ **Normally 10– 20 ml of** fluid is spread in a thin layer b/n the two layers of pleurae.
- ✓ Pleural effusions are classified as **transudates and exudates** based on laboratory analysis of the fluid.

- **Parapneumonic effusions**
 - ✓ Associated with bacterial pneumonia, lung abscess, or bronchiectasis
 - ✓ The most common cause of **exudative effusion**

Transudates

- ✓ Largely due to imbalances in **hydrostatic and oncotic pressures** in the chest
- ✓ Result from movement of **fluid from the peritoneal or retroperitoneal spaces**
- ✓ From **iatrogenic causes**, such as crystalloid infusion into a central venous catheter

Exudates

- ✓ Results primarily from **pleural and lung inflammation** (resulting in a capillary protein leak) or
- ✓ From **impaired lymphatic drainage** of the pleural space (resulting in decreased removal of protein from the pleural space)

- Result from movement of fluid from the peritoneal space, as seen with acute or chronic pancreatitis, and peritoneal carcinomatosis
- Disease in virtually any organ can cause exudative pleural effusions by a variety of mechanisms, including infection, malignancy, immunologic responses, lymphatic abnormalities, noninfectious inflammation, iatrogenic causes

Clinical findings

- ✓ Patients with aerobic bacterial pneumonia and parapneumonic effusions present with an acute febrile illness consisting of chest pain, sputum production, and leukocytosis.
- ✓ Patients with anaerobic infections present with a subacute illness with weight loss, leukocytosis, mild anemia, and a history of some factor that predisposes them to aspiration

- Pleuritic **chest pain and dyspnea** are the most common symptoms, but many pleural effusions are **asymptomatic and discovered** on physical examination or chest x-ray
- Physical examination on the affected side, **absent tactile fremitus, percussion dullness, and decreased or absent breathes sounds**

Diagnosis and laboratory findings

- **Chest x-ray**

- ✓ The most precise way to confirm the physical findings
- ✓ It demonstrates the presence of pleural fluid as **homogenous opacity with a meniscus-sign and obliteration of the costophrenic angle**
- ✓ Large pleural effusions may result in complete **opacification** of the hemithorax and **mediastinal shift to the opposite side**
- The best way to identify and localize a loculated pleural effusion is with **ultrasonography**
- **Pleural fluid** should be performed to confirm the presence of fluid and to determine its characteristics

Diagnostic criteria

- According to Light's criteria, if at least one of the following three criteria is present, the fluid is defined as an exudates
- ✓ Pleural fluid protein/serum protein ratio greater than 0.5
- ✓ Pleural fluid LDH/serum LDH ratio greater than 0.6
- ✓ Pleural fluid LDH greater than two-thirds the upper limits of the laboratory's normal serum LDH

- Proposed **two-criteria and three-criteria diagnostic rules** — which require one criterion to be met to define an exudate include:

Two-test rule

- ✓ Pleural fluid cholesterol greater than **45 mg/dL**
- ✓ Pleural fluid LDH greater than **0.45 times** the upper limit of the laboratory's normal serum LDH

Three-test rule

- ✓ Pleural fluid protein greater than 2.9 g/dL (29 g/L)
- ✓ Pleural fluid cholesterol greater than 45 mg/dL (1.165 mmol/L)
- ✓ Pleural fluid LDH greater than 0.45 times the upper limit of the laboratory's normal serum LDH

- ✓ The possibility of a parapneumonic effusion should be considered whenever a patient with bacterial pneumonia is initially evaluated.
- ✓ The presence of free pleural fluid can be demonstrated with a lateral decubitus radiograph, computed tomography (CT) of the chest, or ultrasound.
- ✓ If the free fluid separates the lung from the chest wall by >10 mm, a therapeutic thoracentesis should be performed

- Factors indicating the likely need for a procedure more invasive than a thoracentesis (in increasing order of importance) include the following:
 - ✓ Loculated pleural fluid
 - ✓ Pleural fluid PH <7.20
 - ✓ Pleural fluid glucose <3.3 mmol/l (<60 mg/dl)
 - ✓ Positive gram stain or culture of the pleural fluid
 - ✓ Presence of gross pus in the pleural space

ANTIBIOTIC THERAPY

- ✓ The choice of antibiotic should target the likely cause of the underlying pneumonia
- ✓ Anaerobic organisms are common, particularly among patients who present with **empyema**
- ✓ Options for empiric therapy that would cover anaerobic organisms include clindamycin, **beta-lactam plus beta-lactamase inhibitors** (eg, amoxicillin-clavulanate)
- ✓ Single agent therapy with either penicillin or metronidazole is not considered optimal.

Note: Some parapneumonic effusions and all empyemas **will need a drainage procedure**

Lung abscess

- **Learning Objective**

At the end of this session, you will be able to

- ✓ Define lung abscess
- ✓ List the etiologies of lung abscess
- ✓ Explain the risk factors for lung abscess
- ✓ Identify the clinical manifestations of lung
- ✓ Make a diagnosis of lung abscess
- ✓ Identify the complications of lung abscess
- ✓ Give treatment for lung abscess
- ✓ Design appropriate methods of prevention of lung abscess

- Definition

- ✓ Lung abscess is defined as collection of pus within a destroyed portion of the lung
- ✓ It may develop associated with the following :-
 - Necrotizing infections of the lung (bacterial pneumonia, TB, fungi)
 - Loss of blood supply to a part of the lung causing cavitory infarction due to septic or embolism, obstruction to airways or cavitations of malignancy
 - The anaerobic abscess is the commonest and usually follows periodontal diseases or aspiration of oropharyngeal /gastric contents

- **Etiology**

- ✓ Lung abscess is commonly caused by **pyogenic bacteria, S. aureus, Klebsiella, mixed anaerobes**
- ✓ Common risk factors include **alcoholism, immunodeficiency, loss of consciousness, and periodontal diseases**

- ✓ Bacteroides oralis group,
- ✓ Prevotella
- ✓ Fusobacterium spp,
- ✓ Peptostreptococcus,
- ✓ Microaerophilic streptococci
- ✓ Aerobic streptococci

- **Clinical features**

- In the **early stage** manifestations of lung abscess may **resemble that of pneumonia**
- The development of features of lung abscess within **1-2 weeks** after bacterial pneumonia, possible aspiration or bronchial obstruction is usually reported
- Patients will have **cough** with sudden **expectoration of massive purulent and foul smelling sputum**, **high grade fever**, and sweating with occasional **hemoptysis**

- **Diagnosis**

- ✓ The diagnosis of lung abscess may not difficult.
- ✓ It should be confirmed by chest x-ray demonstrating parenchymal infiltrates with cavity containing air-fluid level
- ✓ Gram stain and culture of the sputum help to make etiologic diagnosis

- **Medical therapy**

- Antibiotics – start empirical antibiotics until laboratory results are available and adjust drugs accordingly
- Should be given for 4 – 8 weeks.

- ✓ Clindamycin
- ✓ B-lactamases, and clindamycin
- ✓ Metronidazole has done poorly unless combined with a -lactam
- ✓ Vancomycin

- Drainage of abscess
- Surgical treatment of lung abscess may be indicated in patients with
 - ✓ Persistent or massive hemoptysis
 - ✓ Empyema or bronchopleural fistula
 - ✓ Failure of medical treatment

- **Complications**

- ✓ Metastatic brain abscess
- ✓ Fatal hemoptysis
- ✓ Empyema
- ✓ Secondary amyloidosis : - a condition marked by
the accumulation of a protein-based substance

Chronic Obstructive Pulmonary Disease (COPD)

Set By Minichil G

Learning objective

- At the end of this session, you will be able to
 - ✓ Define chronic obstructive pulmonary diseases
 - ✓ List the etiologies of COPD
 - ✓ Describe the pathophysiology of COPD
 - ✓ Identify the clinical manifestations of COPD
 - ✓ Make a diagnosis of COPD
 - ✓ Give therapy to COPD
 - ✓ Design appropriate methods of prevention of COPD

Definition

- ✓ Its pulmonary component is characterized by airflow limitation that is not fully reversible
- ✓ The airflow limitation is usually progressive and associated with an abnormal inflammatory response of the lungs to noxious particles or gases

- Chronic bronchitis
 - ✓ A disease of the airways defined as the **presence of cough and sputum production** for at least a combined **total of 3 months in each of 2 consecutive years**
- Pulmonary emphysema
 - ✓ A pathological term that describes an abnormal **distention of the air spaces** beyond the terminal bronchioles, with **destruction of the walls of the alveoli**.
 - ✓ Impaired gas exchange results from **destruction of the walls of over distended alveoli**

- **Etiology**

- ✓ Smoking
- ✓ Genetic factors (eg, **alpha1 antitrypsin deficiency**)
- ✓ Imbalances of **proteinases and anti proteinases**
- ✓ Prolonged and intense exposure to occupational **dusts and chemicals, indoor and outdoor air pollution**

Pathological changes

■ Chronic bronchitis

- ✓ Hypertrophy of mucus glands
- ✓ Narrowing of airway with thickening of walls
- ✓ Excess production of mucus
- ✓ Pulmonary hypertension and right heart failure in the long term

- **Emphysema**

- ✓ Destruction of alveolar septa and distension of alveoli
- ✓ Mucus production and airway narrowing
- ✓ Reduction in ventilation
- ✓ Less chance of development of pulmonary hypertension and cor-pulmonale

- **Clinical features**

- ✓ Chronic cough
- ✓ Gradual progressive exertional dyspnea
- ✓ Wheezing
- ✓ Recurrent respiratory infections
- ✓ Weakness, weight loss

Physical findings

- ✓ ↑AP diameter
- ✓ Use of accessory muscles of respiration
- ✓ Silent chest
- ✓ Decreased cardiac dullness
- ✓ Hyper- resonant
- ✓ Clubbing nails
- ✓ Sign of anemia

- **Diagnosis method**

- ✓ History / risk factors

- ✓ P/E

- ✓ Chest X- ray

- ✓ Pulmonary function tests

- ✓ Blood gas study

- ✓ Red cell counts may reveal erythrocytosis and elevated hematocrit

Chest x- ray finding

- ✓ Long, narrowed tubular heart
- ✓ Low diaphragm (flat diaphragm)
- ✓ Dark lung fields (translucent) with loss/
increase vascular markings
- ✓ Enlarged heart, large pulmonary artery

Management of COPD

- ✓ So far no curative treatment
- ✓ Therapy is directed at relieving symptoms, controlling potentially fatal exacerbations, and slowing of the progression of the disorder
- ✓ Avoidance of bronchial irritants, especially cessation of smoking

The treatment is outlined as follows

- **Treatment of infection:** COPD patients with purulent sputum should be treated with a broad spectrum antibiotic
- **Control of bronchospasm:** β -agonists like salbutamol
- **Facilitation of drainage of bronchial secretion**
- ✓ Adequate hydration with oral fluids is essential to prevent drying of secretions
- ✓ Inhalation of mist, postural drainage, and chest exercise may help.

- **Hypoxemia:** Oxygen should be given
- **Control of heart failure**
 - ✓ Correction of hypoxemia
 - ✓ Administration of diuretics and restriction of sodium intake
- **Exercise**
 - ✓ Regular exercise as long as there is no severe heart disease its recommended

- **Depression:** Psychological support is very important.
- **Treatment of exacerbation:** exacerbation requires prompt treatment.
- **Phlebotomy:** hematocrit is usually high because of the chronic hypoxemia (polycythemia) especially in patients with predominant chronic bronchitis.

Note

✓ **Smoking cessation** is the single most effective intervention to **prevent COPD or slow its progression**

Complications

- ✓ Respiratory insufficiency and failure are major life-threatening complications of COPD
- ✓ Pneumonia
- ✓ Atelectasis
- ✓ Pneumothorax
- ✓ Cor-pulmonale

Bronchiectasis

- **Def:-** is a chronic & permanent /irreversible **dilatation of the bronchi and bronchioles** due to destruction of bronchial wall.
- **Pathology**
 - ✓ The **dilatation** is due to destructive & Inflammatory changes in the wall of medium-sized air ways.
 - ✓ The normal components of the wall, including **cartilage, muscle, & elastic tissue**, are destroyed & may be replaced by **fibrous tissue**.

Cont---

- ✓ The dilated airways are frequently **contains thick, purulent materials**
- ✓ **Thick sputum obstructs** the bronchi and the wall permanently dilated by sever coughing.

Cont....

✓ Damage bronchia wall



✓ Thick sputum production



✓ Sputum Obstruct bronchi



✓ Wall of bronchi will be distended by cough



✓ Distended air way

Causes

- ✓ Pulmonary infections
- ✓ Obstruction of the bronchus
- ✓ Aspiration of foreign bodies & vomitus
- ✓ Pressure from tumors, dilated blood vessels & enlarged lymph nodes.
- ✓ Idiopathic causes

Predisposing factors

- ✓ Infection in early childhood (measles, influenza, TB, & immuno -deficiency disorders)
- ✓ Surgery (when patient is unable to cough)

Cont---

➤ C/M:

- ✓ Chronic cough with copious amount of purulent sputum (up to 200ml)
- ✓ Hemoptysis (50-70%)
- ✓ on P/E : clubbing of the fingers and cyanosis, crackles, rhonchi, & wheezes may be heard

➤ ***Diagnosis:***

- ✓ The definitive diagnostic evaluation is prolonged hx of productive cough and –ve for PTB.
- ✓ Bronchography & Bronchoscope
- ✓ Chest x ray

Cont---

➤ **Medical management:-**

Objective

- is to prevent and control infection
- Promote bronchial drainage
- Antibiotic based on the results of culture & sensitivity
- Postural drainage
- Bronchodilators may be given

Atelectasis

- Defⁿ It is collapse of an alveolus, a lobule, or larger lung units.

➤ Causes:

- ✓ **Obstruction of a bronchus** by wide variety of diseases e.g. pneumonia , asthma or foreign body, or a plug of thick exudates.
- ✓ **Post operative complication** is the most common cause, the onset is usually **24-72 hrs** after operation,
- ✓ It may also resulted from **pressure in the thorax caused by pleural effusion , pneumothorax** etc.

Cont---

➤ **Risk factors:-**

- ✓ Respiratory depression from opioid, sedatives, & abdominal distention
- ✓ Post operative patients, bedridden patients
- ✓ Pleural effusion, pneumothorax

➤ **C/M:**

- ✓ Marked dyspnea
- ✓ Cyanosis
- ✓ Pleural pain
- ✓ Tachycardia and fever

➤ **Medical Mgt:-**

- Aspirate the plural effusion & pneumothorax
- Force the pt to cough and hyperventilate,
- Mechanical ventilation may be necessary
- Remove the causes /removal of bronchial obstruction

➤ **Prevention:**

- Encouraging the patient to cough, aspirate secretion, & postural drainage.
- Turn the patient frequently.
- Keep postoperative pts in their side.
- Coughing & deep breathing exercise postoperative.

PNEUMOTHORAX

- **Definition**

- ✓ Pneumothorax is the presence of gas in the pleural space

- ✓ **Types of Pneumothorax**

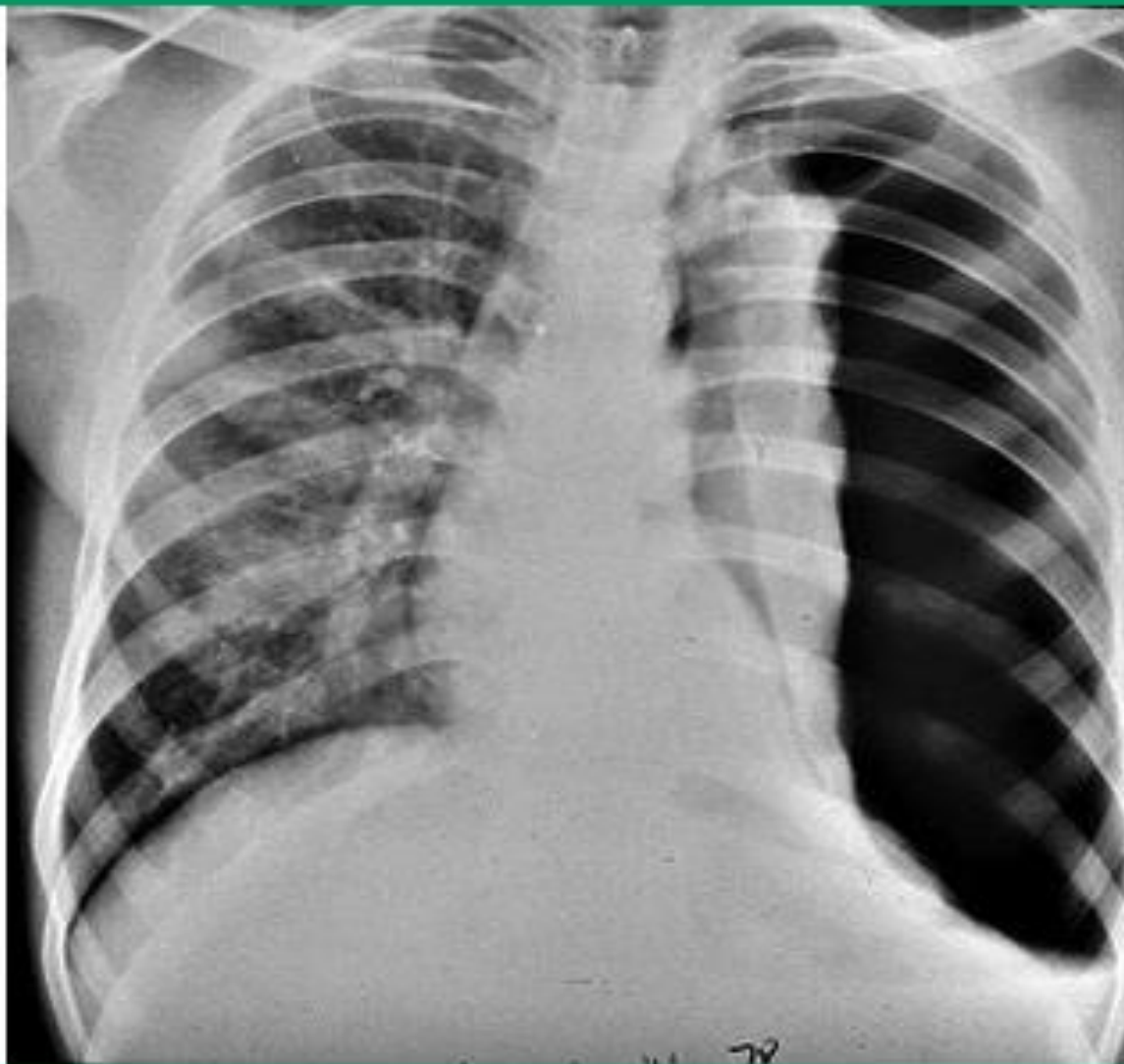
- Primary spontaneous pneumothorax

- Secondary pneumothorax

- A traumatic pneumothorax

- Tension pneumothorax

Tension pneumothorax



Tension pneumothorax due to extensive tuberculosis in the

- **A primary spontaneous pneumothorax (PSP)**
- ✓ A pneumothorax that occurs without a precipitating event in a person or **who does not have known lung disease**
- ✓ Most individuals with PSP have **unrecognized lung disease** with the pneumothorax resulting from **rupture of pleural structure**

- **RISK FACTORS**

- ✓ Smoking

- ✓ Family history???

- **Clinical presentation**

- ✓ Usually occurs when the patient is **at rest**
- ✓ Usually complain of the sudden onset of
dyspnea and pleuritic chest pain
- ✓ Dyspnea being more prominent if the
pneumothorax is large

- Characteristic physical findings
 - ✓ Decreased chest excursion on the affected side
 - ✓ Diminished breath sounds
 - ✓ Hyperresonant percussion
 - ✓ Evidence of labored breathing and hemodynamic compromise (eg, tachycardia, hypotension)
 - ✓ Hypoxemia is common because collapsed and poorly ventilated portions of lung

- **Imaging**

- ✓ Demonstrating a **white visceral pleural line** on the chest radiograph
- ✓ The visceral pleural line defines the **interface of the lung and pleural air**
- ✓ It is either **straight or convex towards** the chest wall
- ✓ **Contralateral** shift of the trachea and mediastinum

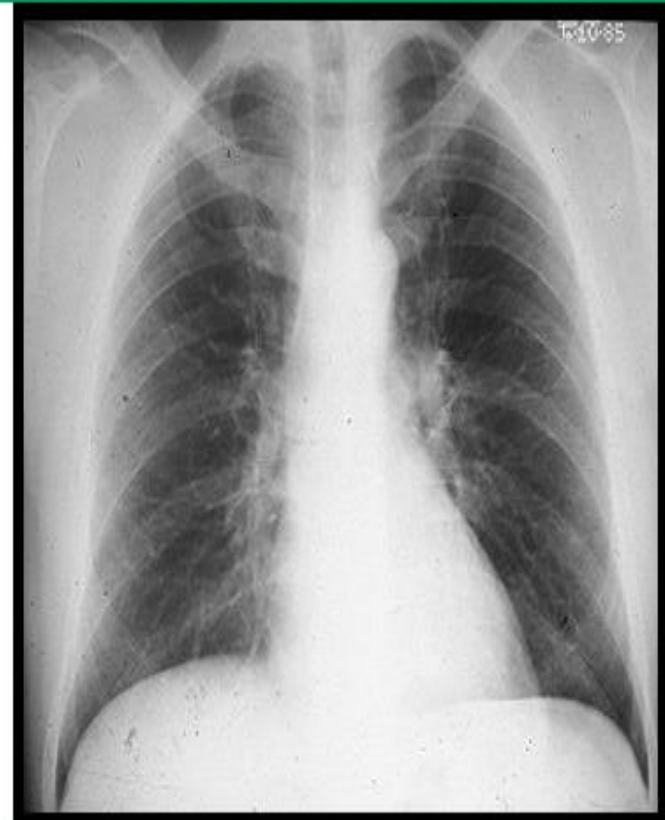
Pneumothorax after stab wound



Chest radiograph shows left apicolateral pneumothorax with typical convex white visceral pleural line.

Courtesy of Paul Chalk, MD.

Normal chest radiograph



Posteroanterior view of a normal chest radiograph.
Courtesy of Carol M Black, MD.

DIAGNOSIS

✓ Chest x-ray

✓ CT scan

- **Management**

- ✓ Smoking cessation

- ✓ Supplemental oxygen

- ✓ Needle aspiration of intrapleural air

- ✓ Chest tube insertion (ie, tube thoracostomy)

■ **A secondary spontaneous pneumothorax (SSP)**

- ✓ Defined as a pneumothorax that occurs as a complication of underlying lung disease

Etiologies

- ✓ Every lung disease can be complicated by (SSP)
 - COPD
 - Pneumocystis jirovecii infection
 - Cystic fibrosis
 - Tuberculosis

Clinical features

- ✓ Dyspnea and chest pain on the same side
- ✓ Decreased chest excursion on the affected side,
- ✓ Diminished breath sounds
- ✓ Hyperresonant percussion
- ✓ Arterial blood gases are frequently abnormal

DIAGNOSIS

- ✓ Chest x-ray
- ✓ CT scan
- ✓ Lung biopsy

- **Management**

- ✓ Initial management is directed at removing air from the pleural space
- ✓ Subsequent management directed at preventing recurrence

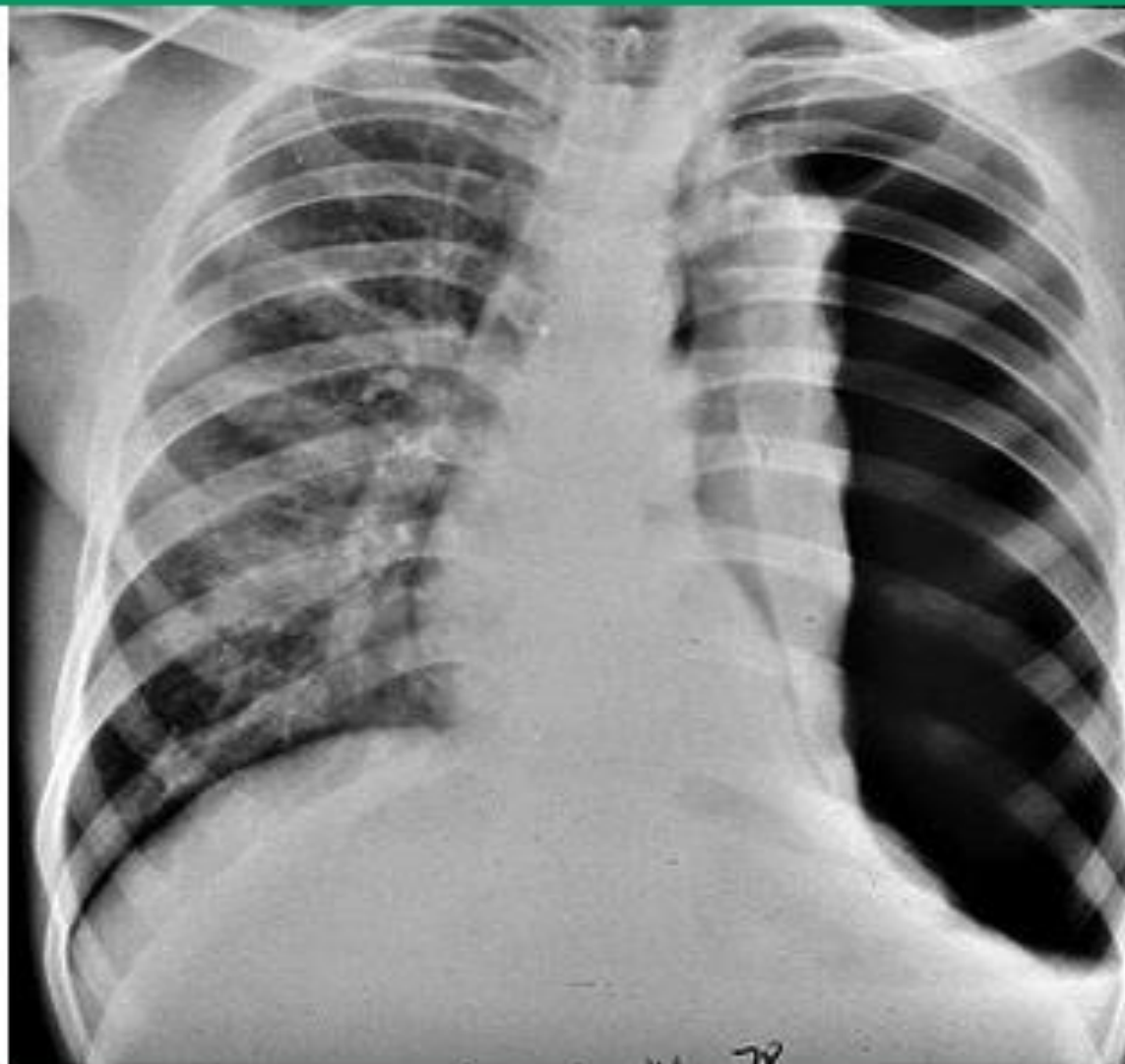
- **Initial management**

- ✓ Patients who are clinically stable and have a small pneumothorax (≤ 2 cm b/n the lung and chest wall)
- ✓ Patients who are clinically stable but have a large pneumothorax should have a chest tube placed (ie, tube thoracostomy)
- ✓ Clinically unstable patients should have a chest tube placed regardless of the size of the pneumothorax
- Tube thoracostomy is preferred over needle aspiration because it is more likely to be successful

Tension Pneumothorax

- This condition usually occurs during mechanical ventilation or resuscitative efforts
- The positive pleural pressure is life-threatening
- ✓ Resulting in decreased venous return to the heart and reduced cardiac output

Tension pneumothorax



Tension pneumothorax due to extensive tuberculosis in the

Traumatic Pneumothorax

- ✓ Can result from both **penetrating and nonpenetrating** chest trauma
- ✓ Should be treated with tube thoracostomy

Pneumothorax after stab wound



Chest radiograph shows left apicolateral pneumothorax with typical convex white visceral pleural line.

Empyema

- ✓ It is a collection of purulent liquid / pus/ in the pleural cavity
- ✓ It may occur **if the lung abscess** extends through pleural cavity

C/M: -Fever, anorexia, & weight loss

–Night sweating ,chest pain , dyspnea

Dx

- ✓ History & physical examination
 - On auscultation absence of breath sounds
 - On percussion –dullness
- ✓ Chest x ray
- ✓ Pleural fluid analysis (AFB, staining, protein, glucose, LDH)

- Medical Mx:–
 - ✓ Thoracentesis: -if fluid is not too thick
 - ✓ Closed –chest drainage tube
 - ✓ Antibiotic
 - Nursing interventions
 - ✓ Resolution of empyema is prolonged process
 - ✓ Instruct breathing exercises
 - ✓ Provide care specific to drainage of pleural fluid

Pulmonary Embolism(PE)

❖ PE is obstruction of the pulmonary artery or one of its branches by a thrombus or any non-blood component that originates somewhere in the venous system or in the right side of the heart.

Risk factors

❖ **Venous stasis** / slowing of blood flow in veins/

which may be due to:-

- Prolonged immobilization / post operative/
- Prolonged period of sitting
- Varicose veins

❖ **Hypercoagulability** / due to release of tissue thromboplastin in often injury / surgery/ due to:-

- ✓ Injury
- ✓ Tumor
- ✓ Increased platelet count

❖ **Venous endothelial disease** such as:--

Thrombophlebitis

Pathophysiology

- ❖ Partial or full obstruction of pulmonary artery by emboli.
- ❖ Reduced or no blood perfusion to the distal part.
- ❖ Regional blood vessels and bronchioles constrict.
- ❖ Increased pulmonary vascular resistance.
- ❖ Cause increase in pulmonary arterial pressure and, in turn, an increase in right ventricular work.
- ❖ When the work requirements of the right ventricle exceed its capacity, **right ventricular failure occurs, leading to a decrease in cardiac output**

Clinical Manifestation

- ❖ May be non specific
- ❖ Chest pain is the commonest symptom
- ❖ Sudden onset of dyspnea is the second most common symptom
- ❖ Tachypnea & Haemoptysis
- ❖ Tachycardia / rapid & weak pulse/
- ❖ Cough & diaphoresis
- ❖ Syncope & sudden death

Cont---

- ❖ N.B. Multiple small emboli can lodge in the terminal pulmonary arterioles, producing multiple small infarctions of the lungs

➤ Diagnostic evaluation

- ❖ Chest x – ray
- ❖ Pulmonary angiography

Management

- **Preventions:-**

- ✓ Prevent deep vein thrombosis / active leg exercise to avoid venous stasis, early ambulation, & use of elastic stockings /.
- ✓ Anticoagulant therapy
- ✓ Avoid leaving IV catheter in veins for prolonged periods.
- ✓ Elevate legs above level of heart
- ✓ Advice against habits that increase venous stasis such as:
 - Crossing leg
 - Sitting or lying down for prolonged period
 - Wearing constricting clothing

- **Emergency interventions**

- ✓ Oxygen administration to relieve hypoxemia, respiratory distress, & cyanosis
- ✓ Open IV infusion
- ✓ Catheterize to monitor urinary output
- ✓ IV diuretics

Medical management

- Anticoagulation therapy
 - ✓ Heparin 5,000 units IV followed by continuous infusion of 1,000 unit per hour is used to prevent recurrence of emboli.
- Thrombolytic therapy (urokinase, streptokinase)
- **Surgical intervention**
 - ✓ Pulmonary embolectomy

Nursing interventions

- ✓ Prevention:-preventing thrombus formation / ambulation, active & passive exercise to prevent venous stasis /
- ✓ Pain management
- ✓ Oxygen administration
- ✓ Patient education
- ✓ Coping with anxiety

- Assignments on common therapeutic and diagnostic modalities